Major Outcomes in High-Risk Hypertensive Patients Randomized to Angiotensin-Converting Enzyme Inhibitor or Calcium Channel Blocker vs Diuretic

The Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT)

The ALLHAT Officers and Coordinators for the ALLHAT Collaborative Research Group

Context Antihypertensive therapy is well established to reduce hypertension-related morbidity and mortality, but the optimal first-step therapy is unknown.

Objective To determine whether treatment with a calcium channel blocker or an angiotensin-converting enzyme inhibitor lowers the incidence of coronary heart disease (CHD) or other cardiovascular disease (CVD) events vs treatment with a diuretic.

Design The Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT), a randomized, double-blind, active-controlled clinical trial conducted from February 1994 through March 2002.

Setting and Participants A total of 33 357 participants aged 55 years or older with hypertension and at least 1 other CHD risk factor from 623 North American centers.

Interventions Participants were randomly assigned to receive chlorthalidone, 12.5 to 25 mg/d (n=15 255); amlodipine, 2.5 to 10 mg/d (n=9048); or lisinopril, 10 to 40 mg/d (n=9054) for planned follow-up of approximately 4 to 8 years.

Main Outcome Measures The primary outcome was combined fatal CHD or nonfatal myocardial infarction, analyzed by intent-to-treat. Secondary outcomes were all-cause mortality, stroke, combined CHD (primary outcome, coronary revascularization, or angina with hospitalization), and combined CVD (combined CHD, stroke, treated angina without hospitalization, heart failure [HF], and peripheral arterial disease).

Results Mean follow-up was 4.9 years. The primary outcome occurred in 2956 participants, with no difference between treatments. Compared with chlorthalidone (6-year rate, 11.5%), the relative risks (RRs) were 0.98 (95% CI, 0.90-1.07) for amlodipine (6-year rate, 11.3%) and 0.99 (95% CI, 0.91-1.08) for lisinopril (6-year rate, 11.4%). Likewise, all-cause mortality did not differ between groups. Five-year systolic blood pressures were significantly higher in the amlodipine (0.8 mm Hg, $P=.03$) and lisinopril (2 mm Hg, $P<.001$) groups compared with chlorthalidone, and 5-year diastolic blood pressure was significantly lower with amlodipine (0.8 mm Hg, $P<.001$). For amlodipine vs chlorthalidone, secondary outcomes were similar except for a higher 6-year rate of HF with amlodipine (10.2% vs 7.7%; RR, 1.38; 95% CI, 1.25-1.52). For lisinopril vs chlorthalidone, lisinopril had higher 6-year rates of combined CVD (33.3% vs 30.9%; RR, 1.10; 95% CI, 1.05-1.16); stroke (6.3% vs 5.6%; RR, 1.15; 95% CI, 1.02-1.30); and HF (8.7% vs 7.7%; RR, 1.19; 95% CI, 1.07-1.31).

Conclusion Thiazide-type diuretics are superior in preventing 1 or more major forms of CVD and are less expensive. They should be preferred for first-step antihypertensive therapy.

JAMA. 2002;288:2981-2997 www.jama.com

Treatment and complications among the 50 to 60 million people in the United States with hypertension are estimated to cost $37 billion annually, with antihypertensive drug costs alone accounting for an estimated $15.5 billion per year.[1] Antihypertensive drug therapy substantially reduces the risk of hypertension-related morbidity and mortality.[2-6] However, the optimal choice for initial pharmacotherapy of hypertension is uncertain.[7]

Earlier clinical trials documented the benefit of lowering blood pressure (BP) using primarily thiazide diuretics or β-blockers.[2,3,8] After these studies, several newer classes of antihypertensive agents (ie, angiotensin-converting enzyme [ACE] inhibitors, calcium channel blockers [CCBs], α-adrenergic blockers, and more recently angiotensin-receptor blockers) became available. Over the past decade, major placebo-controlled trials have documented that ACE inhibitors and CCBs reduce cardiovascular events in individuals with hypertension.[9-11] However, their relative value compared with older, less expensive agents remains unclear. There has been considerable uncertainty regarding effects of some classes of antihypertensive drugs on risk of

See also pp 2998 and 3039.

Author Affiliations: ALLHAT Authors, Their Financial Disclosures, and Group Members are listed at the end of this article.
Corresponding Authors and Reprints: Jackson T. Wright, Jr, MD, PhD, Case Western Reserve University, General Clinical Research Center, Suite 7311, Horvitz Tower, 11000 Euclid Ave, Cleveland, OH 44106-5041 (e-mail: jxw20@po.cwru.edu); Barry R. Davis, MD, PhD, University of Texas-Houston Health Science Center, School of Public Health, 1200 Herman Pressler St, Suite E801, Houston, TX 77030 (e-mail: bdavis@sph.uth.tmc.edu).

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

coronary heart disease (CHD).[6,12-16] The relative benefit of various agents in high-risk hypertensive subgroups such as older patients, black patients, and patients with diabetes also needed to be established.[17]

The Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT), a randomized, double-blind, multicenter clinical trial sponsored by the National Heart, Lung, and Blood Institute, was designed to determine whether the occurrence of fatal CHD or nonfatal myocardial infarction is lower for high-risk patients with hypertension treated with a CCB (represented by amlodipine), an ACE inhibitor (represented by lisinopril), or an α-blocker (represented by doxazosin), each compared with diuretic treatment (represented by chlorthalidone).[18] Chlorthalidone was found to be superior to doxazosin and was previously reported after early termination of the doxazosin arm of the trial.[19,20] Secondary outcomes included all-cause mortality, stroke, and other cardiovascular disease (CVD) events. A lipid-lowering subtrial was designed to determine whether lowering cholesterol with 3-hydroxy-3-methylglutaryl coenzyme A reductase inhibitor (pravastatin) compared with usual care reduced all-cause mortality in a moderately hypercholesterolemic subset of ALLHAT participants.[18,21] To evaluate differences in CVD effects of the various first-step drugs, ALLHAT was designed with a large sample size (9000-15 000 participants/intervention arm) and long follow-up (4-8 years). This study presents results of the amlodipine and lisinopril vs chlorthalidone comparisons on major CVD outcomes.

METHODS

Study Design

The rationale and design of ALLHAT have been presented elsewhere.[18] Participants were men and women aged 55 years or older who had stage 1 or stage 2 hypertension with at least 1 additional risk factor for CHD events.[18,22] The risk factors included previous (>6 months) myocardial infarction or stroke, left ventricular hypertrophy demonstrated by electrocardiography or echocardiography, history of type 2 diabetes, current cigarette smoking, high-density lipoprotein cholesterol of less than 35 mg/dL (<0.91 mmol/L), or documentation of other atherosclerotic CVD. Individuals with a history of hospitalized or treated symptomatic heart failure (HF) and/or known left ventricular ejection fraction of less than 35% were excluded.

Unless the drug regimen had to be tapered for safety reasons, individuals continued any prior antihypertensive medications until they received randomized study drug, at which point they stopped taking all previous medications. Treatment with the study drug was initiated the day after randomization. By telephone, participants were randomly assigned to chlorthalidone, amlodipine, or lisinopril in a ratio of 1.7:1:1. The concealed randomization scheme was generated by computer, implemented at the clinical trials center, stratified by center and blocked in random block sizes of 5 or 9 to maintain balance. Participants (n=33 357) were recruited at 623 centers in the United States, Canada, Puerto Rico, and the US Virgin Islands between February 1994 and January 1998. (The original reported number of 625 sites changed because 2 sites and their patients with poor documentation of informed consent were excluded.[20]) All participants gave written informed consent, and all centers obtained institutional review board approval. Follow-up visits were at 1 month; 3, 6, 9, and 12 months; and every 4 months thereafter. The range of possible follow-up was 3 years 8 months to 8 years 1 month. The closeout phase began on October 1, 2001, and ended on March 31, 2002.

Treatment

Trained observers using standardized techniques measured BPs during the trial.[20] Visit BP was the average of 2 seated measurements. Goal BP in each randomized group was less than 140/90 mm Hg achieved by titrating the assigned study drug (step 1) and adding open-label agents (step 2 or 3) when necessary. The choice of step 2 drugs (atenolol, clonidine, or reserpine) was at the physician's discretion. Nonpharmacologic approaches to treatment of hypertension were recommended according to national guidelines.[4,23] Step 1 drugs were encapsulated and identical in appearance so that the identity of each agent was double-masked at each dosage level. Dosages were 12.5, 12.5 (sham titration), and 25 mg/d for chlorthalidone; 2.5, 5, and 10 mg/d for amlodipine; and 10, 20, and 40 mg/d for lisinopril. Doses of study-supplied open-label step 2 drugs were 25 to 100 mg/d of atenolol; 0.05 to 0.2 mg/d of reserpine; or 0.1 to 0.3 mg twice a day of clonidine; step 3 was 25 to 100 mg twice a day of hydralazine. Other drugs, including low doses of open-label step 1 drug classes, were permitted if clinically indicated.[18,20]

Outcomes

The primary outcome was fatal CHD or nonfatal myocardial infarction combined.[18] Four major prespecified secondary outcomes were all-cause mortality, fatal and nonfatal stroke, combined CHD (the primary outcome, coronary revascularization, hospitalized angina), and combined CVD (combined CHD, stroke, other treated angina, HF [fatal, hospitalized, or treated nonhospitalized], and peripheral arterial disease). Coronary revascularization included coronary artery bypass graft, percutaneous angioplasty, insertion of stents, and atherectomy. Individual components of the combined outcomes were prespecified and examined, as were other secondary outcomes including cancer, incident electrocardiographic left ventricular hypertrophy, end-stage renal disease (ESRD) (dialysis, renal transplant, or death), and slope of the reciprocal of longitudinal serum creatinine measurements. Change in estimated glomerular filtration rate[24,25] was examined post hoc.

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

Study outcomes were assessed at follow-up visits and reported to the clinical trials center.[18] Hospitalized outcomes were primarily based on clinic investigator reports, and copies of death certificates and hospital discharge summaries were requested. Among all combined CVD events that resulted in deaths, hospitalizations, or both, the proportion with documentation (ie, a death certificate or a hospital discharge summary) was 99% in all 3 treatment groups. In addition, searches for outcomes were accomplished through the Center for Medicare and Medicaid Services, the Department of Veterans Affairs, the National Death Index, and the Social Security Administration databases. A death was ascertained by clinic report or by match with the aforementioned databases plus a confirmatory death certificate. A death pending confirmation is one found using databases but for which a confirmatory death certificate has not yet been obtained. Medical reviewers from the clinical trials center verified the physician-assigned diagnoses of outcomes using death certificates and hospital discharge summaries. More detailed information was collected on a random (10%) subset of CHD and stroke events to validate the procedure of using physician diagnoses.[18] When a large excess of HF became evident in the doxazosin arm, a 1-time sample of HF hospitalizations was reviewed by the ALLHAT Endpoints Subcommittee. Agreement rates between the subcommittee and clinic investigators were 90% (155/172) for the primary outcome, 85% (33/39) for HF hospitalizations,[26] and 84% (129/153) for stroke, and were similar in all treatment groups.

Two major safety outcomes, angioedema and hospitalization for gastrointestinal bleeding, were prespecified. Occurrence of gastrointestinal bleeding was ascertained from Center for Medicare and Medicaid Services and Department of Veterans Affairs hospitalization databases, representing 74% of ALLHAT participants (persons ≥65 years, Department of Veterans Affairs participants, or both).[27] Angioedema was ascertained using a solicited event question on a serious adverse event form.

Statistical Methods

To maximize statistical power, 1.7 times as many participants were assigned to the diuretic group as to each of the other 3 groups.[18] Given the achieved sample size and expected event rate, treatment crossovers, and losses to follow-up, ALLHAT had 83% power to detect a 16% reduction in risk of the primary outcome between chlorthalidone and each other group at a 2-sided $\alpha = .0178$ ($z = 2.37$) to account for the 3 original comparisons.[28] Data were analyzed according to participants' randomized treatment assignments regardless of their subsequent medications (intent-to-treat analysis). Cumulative event rates were calculated using the Kaplan-Meier method. Although rates are presented only through 6 years, both the log-rank test and Cox proportional hazards regression model incorporated the participant's entire trial experience to evaluate differences between cumulative event curves and to obtain 2-sided *P* values. Only the Cox proportional hazard regression results are presented, because *P* values were essentially identical. Hazard ratios (relative risks [RRs]) and 95% confidence intervals (CIs) were obtained from the Cox proportional hazards regression model.[29] For consistency with $\alpha = .0178$, 95% CIs may be converted to 98.2% limits by multiplying the upper limit and dividing the lower limit by $RR^{(0.41/Z)}$, where Z is the value of the test statistic for the RR estimate. The Cox proportional hazards regression model assumption was examined by using log-log plots and testing a treatment × time (time-dependent) interaction term; if it was violated, the RR estimate from a 2-by-2 table was used.[29] Heterogeneity of effects in prespecified subgroups, (1) men and women, (2) participants less than 65 and 65 years or older, (3) black and nonblack participants, and (4) diabetic and nondiabetic participants, and the post hoc subgroups presence or absence of CHD at baseline, was examined by testing for treatment-covariate interaction with the Cox proportional hazards regression model by using $P < .05$. SAS version 8.0 (SAS Institute, Cary, NC) and STATA version 7 (Stata Corp, College Station, Tex) were used for statistical analyses.

A National Heart, Lung, and Blood Institute–appointed data and safety monitoring board met at least annually to review the accumulating data and to monitor for safety and efficacy. The Lan-DeMets version of the O'Brien-Fleming group sequential boundaries was used to assess treatment group differences, and conditional power was used to assess futility.[30,31]

RESULTS

Patient Characteristics

TABLE 1 presents baseline characteristics for the 33 357 participants in the chlorthalidone, amlodipine, and lisinopril treatment groups. The mean age was 67 years; 47% were women, 35% were black, 19% were Hispanic, and 36% were diabetic. There were nearly identical distributions of baseline factors in the 3 treatment groups.[22]

Visit and Medication Adherence

FIGURE 1 shows the number of participants randomized and followed up to the time of closeout. In all 3 treatment groups, the mean (SD) length of follow-up was 4.9 years (1.4 years), and 99% of expected person-years were observed. The maximum range of follow-up was 8.0, 7.9, and 8.1 years in the chlorthalidone, amlodipine, and lisinopril groups, respectively. At trial closeout, 419 (2.7%) of the chlorthalidone group, 258 (2.8%) of the amlodipine group, and 276 (3.0%) of the lisinopril group had unknown vital status. Among participants with unknown vital status, the distributions of most baseline factors were similar among the 3 treatment groups, but participants assigned to lisinopril were less likely to be black and more likely to be women, have untreated hypertension, evidence of CHD or atherosclerotic CVD, and a lower mean serum glucose.

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

Visit adherence decreased over time from about 92% at 1 year to 84% to 87% at 5 years in all 3 treatment groups (TABLE 2). Among participants in the chlorthalidone group who were contacted in the clinic or by telephone within 12 months of annual scheduled visits, 87.1% were taking chlorthalidone or another diuretic at 1 year, decreasing to 80.5% at 5 years; 67.5% (n=4387) were taking a diuretic without a CCB or an ACE inhibitor; and 13.2% were taking a diuretic with a CCB (5.8% [n=399]) or an ACE inhibitor (9.3% [n=641]). Only 9.0% were taking either a CCB (5.8% [n=399]) or an ACE inhibitor (5.6% [n=385]) without a diuretic at 5 years.

Among participants in the amlodipine group, 87.6% were taking amlodipine or another CCB at 1 year, decreasing to 80.4% at 5 years; and 63.8% (n=2502) were taking a CCB alone without a diuretic. Another 16.6% were taking a CCB with a diuretic, and only 6.9% were taking a diuretic without a CCB. Among participants in the lisinopril group, 82.4% were taking lisinopril or another ACE inhibitor at 1 year, decreasing to 72.6% at 5 years; 56.9% (n=2143) were taking an ACE inhibitor alone without a diuretic; and 15.7% were taking an ACE inhibitor with a diuretic at 5 years. About 8.5% were taking a diuretic without an ACE inhibitor.

The most common reasons for not taking step 1 medication at 5 years in the chlorthalidone, amlodipine, and lisinopril groups were unspecified refusals (41.4% [n=775], 40.5% [n=443], and 37.9% [n=552], respectively) and symptomatic adverse effects (15.0% [n=282], 16.4% [n=180], and 18.1% [n=264], respectively). Elevated BP (4.5% [n=84], 3.5% [n=38], and 9.0% [n=131]) or other adverse effects such as abnormal laboratory values (3.8% [n=71], 1.6% [n=17], and 2.3% [n=34]) were other reasons given for discontinuation of step 1 medications. Among participants with available medication information at 1 year, 26.7%, 25.9%, and 32.6% of those assigned to chlorthalidone, amlodipine, and lisinopril, respectively, were taking a step 2 or step 3 drug. At 5 years, the corresponding percentages were 40.7%, 39.5%, and 43.0%, respectively. Usage patterns of specific step 2 drugs were similar among groups. Participants could be taking more than 1 step-up drug. At 1 year, 40.0% (n=4645), 44.0% (n=3017), and 43.8% (n=2764) of participants assigned to chlorthalidone, amlodipine, and lisinopril, respectively, still taking their

Table 1. Baseline Characteristics of the ALLHAT Participants*

Characteristic	No. of Participants (%)		
	Chlorthalidone (n = 15 255)	Amlodipine (n = 9048)	Lisinopril (n = 9054)
Age, mean (SD), y	66.9 (7.7)	66.9 (7.7)	66.9 (7.7)
Age range, y			
55-64	6471 (42.4)	3844 (42.5)	3869 (42.7)
≥65	8784 (57.6)	5204 (57.5)	5185 (57.3)
Ethnicity			
White, non-Hispanic	7202 (47.2)	4305 (47.6)	4262 (47.1)
Black, non-Hispanic	4871 (31.9)	2911 (32.2)	2920 (32.3)
White Hispanic	1912 (12.5)	1108 (12.2)	1136 (12.5)
Black Hispanic	498 (3.3)	302 (3.3)	290 (3.2)
Other	772 (5.1)	422 (4.7)	446 (4.9)
Women	7171 (47.0)	4280 (47.3)	4187 (46.2)
Education, mean (SD), y	11.0 (4.0)	11.0 (3.9)	11.0 (4.1)
Receiving antihypertensive treatment	13 754 (90.2)	8171 (90.3)	8164 (90.2)
Blood pressure, mean (SD), mm Hg	146 (16)/84 (10)	146 (16)/84 (10)	146 (16)/84 (10)
Treated at baseline	145 (16)/83 (10)	145 (16)/83 (10)	145 (16)/84 (10)
Untreated at baseline	156 (12)/89 (9)	157 (12)/90 (9)	156 (12)/89 (9)
Eligibility risk factors†			
Cigarette smoker	3342 (21.9)	1980 (21.9)	1981 (21.9)
Atherosclerotic CVD‡	7900 (51.8)	4614 (51.0)	4684 (51.7)
History of MI or stroke	3581 (23.5)	2098 (23.2)	2058 (22.7)
History of coronary revascularization	1986 (13.0)	1106 (12.2)	1218 (13.5)
Other atherosclerotic CVD	3604 (23.6)	2145 (23.7)	2152 (23.8)
Major ST depression or T-wave inversion	1572 (10.4)	908 (10.1)	940 (10.5)
Type 2 diabetes	5528 (36.2)	3323 (36.7)	3212 (35.5)
HDL-C <35 mg/dL	1798 (11.8)	1018 (11.3)	1061 (11.7)
LVH by electrocardiogram	2467 (16.2)	1533 (16.9)	1474 (16.3)
LVH by echocardiogram	695 (4.6)	411 (4.6)	402 (4.5)
History of CHD at baseline§	3943 (26.0)	2202 (24.5)	2270 (25.3)
Body mass index, mean (SD)	29.7 (6.2)	29.8 (6.3)	29.8 (6.2)
Current medication use			
Aspirin	5426 (35.6)	3268 (36.1)	3258 (36.0)
Estrogen supplementation (women only)	1273 (17.8)	752 (17.6)	727 (17.4)
Lipid trial participants	3755 (24.6)	2240 (24.8)	2167 (23.9)

*ALLHAT indicates Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial; CVD, cardiovascular disease; MI, myocardial infarction; HDL-C, high-density lipoprotein cholesterol; LVH, left ventricular hypertrophy; and CHD, coronary heart disease. Body mass index was calculated as weight in kilograms divided by the square of height in meters. To convert HDL-C to mmol/L, multiply by 0.0259.

†For trial eligibility, participants had to have at least 1 other risk factor in addition to hypertension. Thus, the indicated risk factors are not mutually exclusive or exhaustive and may not represent prevalence.

‡History of MI or stroke, history of coronary revascularization, major ST segment depression or T-wave inversion on any electrocardiogram in the past 2 years, other atherosclerotic CVD (history of angina pectoris; history of intermittent claudication, gangrene, or ischemic ulcers; history of transient ischemic attack; coronary, peripheral vascular, or carotid stenosis ≥50% documented by angiography or Doppler studies; ischemic heart disease documented by reversible or fixed ischemia on stress thalium or dipyridamole thalium, ST depression ≥1 mm for ≥1 minute on exercise testing or Holter monitoring; reversible wall motion abnormality on stress echocardiogram; ankle-arm index <0.9; abdominal aortic aneurysm detected by ultrasonography, computed tomography scan, or radiograph; carotid or femoral bruits).

§P = .03 for comparison of groups.

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

blinded medication were receiving the maximal study dose. At 5 years, the percentages were 56.9% (n=2629), 65.7% (n=1856), and 60.3% (n=1391), respectively.

Intermediate Outcomes

Given the large sample size in ALLHAT, almost all differences in follow-up BP and biochemical measurements were statistically significant (TABLE 3 and TABLE 4). Mean seated BP at randomization was about 146/84 mm Hg in all 3 groups, with 90% of participants reporting current antihypertensive drug treatment (Table 1). Follow-up BPs in all 3 groups are shown in Table 3 and FIGURE 2.

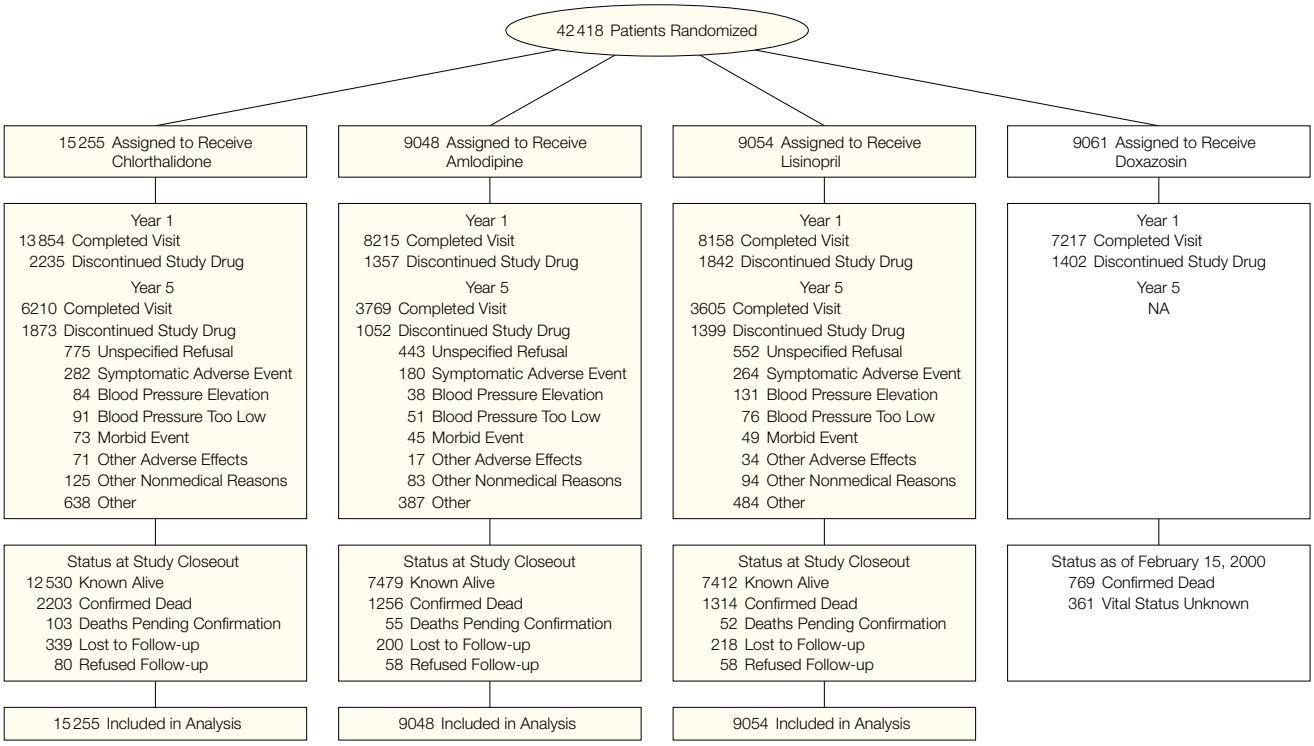
Mean total serum cholesterol levels at baseline and 4 years follow-up are shown in Table 4. At 4 years, about 35% to 36% of participants in all 3 groups reported taking lipid-lowering drugs, largely statins, some as a result of participation in the ALLHAT lipid trial. Mean serum potassium levels at baseline and follow-up are also shown; about 8% of the chlorthalidone group were receiving potassium supplementation at 5 years compared with 4% in the amlodipine group and 2% in the lisinopril group. Among individuals classified as nondiabetic at baseline, with baseline fasting serum glucose less than 126 mg/dL (7.0 mmol/L), incidence of diabetes (fasting serum glucose, ≥126 mg/dL [7.0 mmol/L]) at 4 years was 11.6%, 9.8%, and 8.1%, respectively.

Mean estimated glomerular filtration rate at baseline was about 78 mL/min per 1.73 m^2 in all groups. At 4 years, it was 70.0, 75.1, and 70.7 mL/min per 1.73 m^2 in the chlorthalidone, amlodipine, and lisinopril groups, respectively. The slopes of the reciprocal of serum creatinine over time were virtually identical in the chlorthalidone and lisinopril groups (–0.018 and –0.019 dL/mg per year), whereas the decline in the amlodipine slope (–0.012 dL/mg per year) was less than that of the chlorthalidone slope ($P<.001$).

Primary and Secondary Outcomes

Amlodipine vs Chlorthalidone. No significant difference was observed between amlodipine and chlorthalidone for the primary outcome (RR, 0.98; 95% CI, 0.90-1.07) or for the secondary outcomes of all-cause mortality, combined CHD, stroke, combined CVD, angina, coronary revascularization, peripheral arterial disease, cancer, or ESRD (TABLE 5, FIGURE 3, and FIGURE 4). The amlodipine group had a 38% higher risk of HF ($P<.001$) with

Figure 1. Randomization and Follow-up of Participants in the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial



NA indicates not applicable. Eligibility data were not collected for nonrandomized screenees. All randomized participants were included in the analyses. A patient may have more than 1 reason for discontinuing study drug; therefore, numbers do not sum to total. On January 24, 2000, the National Heart, Lung, and Blood Institute decided to discontinue the doxazosin group and report results.[18] Study closeout for chlorthalidone, amlodipine, and lisinopril groups was from October 1, 2001, through March 31, 2002. Collection of last events for the doxazosin group had a closeout interval from October 15, 1999, through February 15, 2000, which captures more information than that reported previously.[24]

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

a 6-year absolute risk difference of 2.5% and a 35% higher risk of hospitalized/fatal HF ($P<.001$). The treatment effects for all outcomes were consistent across the predefined subgroups (FIGURE 5) and by absence or presence of CHD at baseline. Cause-specific mortality rates (except for unintentional injuries/suicides/homicides in amlodipine compared with chlorthalidone, not a prespecified hypothesis) were similar for the 2 groups (TABLE 6).

Lisinopril vs Chlorthalidone. No significant difference was observed between lisinopril and chlorthalidone for the primary outcome (RR, 0.99; 95% CI, 0.91-1.08) or for the secondary outcomes of all-cause mortality, combined CHD, peripheral arterial disease, cancer, or ESRD (Table 5, Figures 3 and 4). Cause-specific mortality rates were also similar in the 2 groups (Table 6). The lisinopril group had a 15% higher risk for stroke ($P=.02$) and a 10% higher risk of combined CVD ($P<.001$), with a 6-year absolute risk difference for combined CVD of 2.4%. Included in this analysis was a 19% higher risk of HF ($P<.001$), a 10% higher risk of hospitalized/fatal HF ($P=.11$), an 11% higher risk of hospitalized/treated angina ($P=.01$), and a 10% higher risk of coronary revascularization ($P=.05$). The treatment effects for all outcomes were consistent across subgroups by sex, diabetic status (FIGURE 6), and baseline CHD status. For combined CHD, there was a significant differential effect by age ($P=.01$ for interaction) with RRs (lisinopril vs chlorthalidone) of 0.94 for those less than 65 years vs 1.11 in those 65 years or older. However, when age was modeled as a continuous variable, there was no significant interaction. For stroke and combined CVD, there was a significant differential effect by race ($P=.01$ and $P=.04$ for interaction, respectively). The RRs (lisinopril vs chlorthalidone) were 1.40 (95% CI, 1.17-1.68) and 1.00 (95% CI, 0.85-1.17) for stroke and 1.19 (95%

Table 2. Visits Expected and Completed and Antihypertensive Medication Use at Annual Visits

	Years, No. (%)				
	1	2	3	4	5
Chlorthalidone					
Expected visits	15 067 (98.8)	14 711 (96.4)	14 272 (93.6)	12 380 (81.2)	7243 (47.5)
Completed visits	13 854 (91.9)	12 988 (88.3)	12 335 (86.4)	10 618 (85.8)	6210 (85.7)
Receiving blinded study drug	11 618 (83.9)	10 367 (79.8)	9372 (76.0)	8149 (72.9)	4623 (71.2)
Receiving blinded study drug or same class	12 063 (87.1)	11 001 (84.7)	10 202 (82.7)	9034 (80.8)	5247 (80.5)
Full crossovers*	707 (5.1)	865 (6.7)	944 (7.7)	921 (8.2)	583 (9.0)
Partial crossovers†	469 (3.4)	770 (5.9)	1054 (8.5)	1223 (10.9)	860 (13.2)
Receiving step 2 or 3‡	3703 (26.7)	4185 (32.2)	4395 (35.6)	4244 (38.0)	2642 (40.7)
Other antihypertensive medication	594 (4.3)	586 (4.5)	618 (5.0)	609 (5.5)	320 (4.9)
No. of antihypertensive medications, mean (SD)	1.4 (0.7)	1.5 (0.8)	1.6 (0.9)	1.7 (1.0)	1.8 (1.0)
Amlodipine					
Expected visits	8937 (98.8)	8733 (96.5)	8510 (94.0)	7411 (81.9)	4343 (48.0)
Completed visits	8215 (91.9)	7672 (87.9)	7355 (86.4)	6341 (85.6)	3769 (86.8)
Receiving blinded study drug	6858 (83.5)	6106 (79.6)	5630 (76.6)	4886 (73.3)	2826 (72.1)
Receiving blinded study drug or same class	7192 (87.6)	6532 (85.2)	6116 (83.2)	5367 (80.5)	3151 (80.4)
Full crossovers*	232 (2.8)	342 (4.5)	369 (5.0)	401 (6.0)	270 (6.9)
Partial crossovers†	548 (6.7)	666 (8.7)	818 (11.1)	881 (13.2)	649 (16.6)
Receiving step 2 or 3‡	2124 (25.9)	2456 (32.0)	2590 (35.2)	2457 (36.9)	1548 (39.5)
Other antihypertensive medication	478 (5.8)	499 (6.5)	546 (7.4)	568 (8.5)	314 (8.0)
No. of antihypertensive medications, mean (SD)	1.4 (0.7)	1.5 (0.8)	1.7 (0.9)	1.7 (1.0)	1.9 (1.0)
Lisinopril					
Expected visits	8942 (98.8)	8725 (96.4)	8458 (93.4)	7356 (81.2)	4315 (47.7)
Completed visits	8158 (91.2)	7574 (86.8)	7185 (84.9)	6142 (83.5)	3605 (83.5)
Receiving blinded study drug	6316 (77.4)	5418 (71.5)	4897 (68.2)	4155 (64.4)	2307 (61.2)
Receiving blinded study drug or same class	6721 (82.4)	5944 (78.4)	5536 (77.1)	4824 (74.8)	2736 (72.6)
Full crossovers*	285 (3.5)	387 (5.1)	430 (6.0)	455 (7.0)	320 (8.5)
Partial crossovers†	475 (5.8)	662 (8.7)	797 (11.1)	857 (13.3)	593 (15.7)
Receiving step 2 or 3‡	2661 (32.6)	2747 (36.3)	2788 (38.8)	2625 (40.7)	1620 (43.0)
Other antihypertensive medication	836 (10.2)	869 (11.5)	858 (12.0)	822 (12.7)	480 (12.7)
No. of antihypertensive medications, mean (SD)	1.5 (0.8)	1.7 (1.0)	1.8 (1.0)	1.9 (1.1)	2.0 (1.2)

*Full crossovers: (1) assigned to chlorthalidone, not on step 1, no open-label diuretic, but on open-label calcium channel blocker (CCB) or angiotensin-converting enzyme (ACE) inhibitor; (2) assigned to lisinopril, not on step 1, no open-label ACE inhibitor, but on open-label diuretic; (3) assigned to amlodipine, not on step 1, no open-label CCB, but on open-label diuretic.
†Partial crossovers: (1) assigned to chlorthalidone, on step 1 or open-label diuretic and on open-label CCB or ACE inhibitor; (2) assigned to lisinopril, on step 1 or open-label ACE inhibitor and on open-label diuretic; (3) assigned to amlodipine, on step 1 or open-label CCB and on open-label diuretic.
‡Step 2: atenolol, clonidine, or reserpine; step 3: hydralazine.

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

CI, 1.09-1.30) and 1.06 (95% CI, 1.00-1.13) for combined CVD in blacks and nonblacks, respectively.

The mean follow-up systolic BP for all participants was 2 mm Hg higher in the lisinopril group than the chlorthalidone group, 4 mm Hg higher in blacks, and 3 mm Hg higher in those 65 years or older. Adjustment for follow-up BP as time-dependent covariates in a Cox proportional hazards regression model slightly reduced the RRs for stroke (1.15 to 1.12) and HF (1.20 to 1.17) overall and in the black subgroup (stroke, 1.40 to 1.35; and HF, 1.32 to 1.26), but the results remained statistically significant.

Primary Safety Outcomes

Six-year rates of hospitalization for gastrointestinal bleeding, available only in Medicare and Department of Veterans Affairs participants, occurred in 8.8%, 8.0%, and 9.6% participants in the chlorthalidone, amlodipine, and lisinopril treatment groups, respectively, with no significant differences (Table 5). Angioedema occurred in 8 of 15 255 (0.1%), 3 of 9048 (<0.1%), and 38 of 9054 (0.4%) persons in the chlorthalidone, amlodipine, and lisinopril treatment groups, respectively. Significant differences were seen for the lisinopril vs chlorthalidone comparison overall ($P<.001$), in blacks (2 of 5369 [<0.1%] for chlorthalidone, 23 of 3210 [0.7%] for lisinopril; $P<.001$), and in nonblacks (6 of 9886 [0.1%] for chlorthalidone, 15 of 5844 for lisinopril [0.3%]; $P=.002$). The only death from angioedema was in the lisinopril group.

COMMENT

Neither amlodipine (representing CCBs, particularly dihydropyridine [DHP]–CCBs) nor lisinopril (representing ACE inhibitors) was superior to chlorthalidone (representing thiazide-type diuretics) in preventing major coronary events or in increasing survival. Chlorthalidone was superior to amlodipine (by about 25%) in preventing HF, overall, and for hospitalized or fatal cases, although it did not differ from amlodipine in overall CVD prevention. Chlorthalidone was superior to lisinopril in lowering BP and in preventing aggregate cardiovascular events, principally stroke, HF, angina, and coronary revascularization. ALLHAT previously reported that chlorthalidone was superior to doxazosin (representing α-blockers) in reducing BP and preventing cardiovascular events, particularly HF.[19,20]

It is not surprising that no significant differences in CHD and stroke rates were found between chlorthalidone and

Table 3. Number of Participants, Mean Blood Pressure, Achieved Blood Pressure Goal, and Blood Pressure Difference at Baseline and Annual Visits

				P Value	
	Chlorthalidone	Amlodipine	Lisinopril	Amlodipine vs Chlorthalidone	Lisinopril vs Chlorthalidone
			No. of Participants (%)		
Baseline	15 255 (100)	9048 (100)	9054 (100)		
1 Year	12 862 (84.3)	7609 (84.1)	7521 (83.1)		
2 Years	11 740 (77.0)	6883 (76.1)	6700 (74.0)		
3 Years	10 698 (70.1)	6381 (70.5)	6076 (67.1)		
4 Years	9379 (61.5)	5637 (62.3)	5325 (58.8)		
5 Years	5301 (34.7)	3195 (35.3)	2963 (32.7)		
			Systolic Blood Pressure, Mean (SD), mm Hg		
Baseline	146.2 (15.7)	146.2 (15.7)	146.4 (15.7)	.98	.39
1 Year	136.9 (15.8)	138.5 (14.9)	140.0 (18.5)	<.001	<.001
2 Years	135.9 (15.9)	137.1 (15.0)	138.4 (17.9)	<.001	<.001
3 Years	134.8 (15.4)	135.6 (15.2)	136.7 (17.3)	.001	<.001
4 Years	133.9 (15.7)	134.8 (15.0)	135.5 (17.2)	.002	<.001
5 Years	133.9 (15.2)	134.7 (14.9)	135.9 (17.9)	.03	<.001
			Diastolic Blood Pressure, Mean (SD), mm Hg		
Baseline	84.0 (10.1)	83.9 (10.2)	84.1 (10.0)	.52	.49
1 Year	79.3 (9.9)	78.7 (9.5)	79.9 (10.5)	<.001	<.001
2 Years	78.3 (9.6)	77.7 (9.6)	78.6 (10.3)	<.001	.03
3 Years	77.2 (9.5)	76.4 (9.6)	77.3 (10.3)	<.001	.42
4 Years	76.5 (9.7)	75.7 (9.6)	76.6 (10.4)	<.001	.48
5 Years	75.4 (9.8)	74.6 (9.9)	75.4 (10.7)	<.001	.94
			Achieved Blood Pressure Goal of <140/90 mm Hg, No. (%)		
Baseline	4149 (27.2)	2497 (27.6)	2381 (26.3)	.56	.12
1 Year	7434 (57.8)	4200 (55.2)	3806 (50.6)	<.001	<.001
2 Years	7161 (61.0)	3951 (57.4)	3625 (54.1)	<.001	<.001
3 Years	6836 (63.9)	4046 (63.4)	3597 (59.2)	.54	<.001
4 Years	6293 (67.1)	3709 (65.8)	3360 (63.1)	.15	<.001
5 Years	3615 (68.2)	2118 (66.3)	1813 (61.2)	.09	<.001
			Systolic Blood Pressure, Δ mm Hg*		
Baseline		0	0.2		
1 Year		1.6	3.1		
2 Years		1.2	2.5		
3 Years		0.8	1.9		
4 Years		0.9	1.6		
5 Years		0.8	2.0		
			Diastolic Blood Pressure, Δ mm Hg*		
Baseline		−0.1	0.1		
1 Year		−0.6	0.6		
2 Years		−0.6	0.3		
3 Years		−0.8	0.1		
4 Years		−0.8	0.1		
5 Years		−0.8	0		

*Compared with chlorthalidone group.

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

Table 4. Biochemical Changes by Treatment Group*

	Chlorthalidone	Amlodipine	Lisinopril	*P* Value: Amlodipine vs Chlorthalidone	*P* Value: Lisinopril vs Chlorthalidone
Cholesterol, mg/dL					
No. of participants (%)					
Baseline	14 483 (94.9)	8586 (94.9)	8573 (94.7)		
2 Years	10 206 (66.9)	6025 (66.6)	5739 (63.4)		
4 Years	8495 (55.7)	5025 (55.5)	4711 (52.0)		
Mean (SD)					
Baseline	216.1 (43.8)	216.5 (44.1)	215.6 (42.4)	.47	.38
2 Years	205.3 (42.1)	202.5 (42.2)	202.0 (42.8)	<.001	<.001
4 Years	197.2 (42.1)	195.6 (41.0)	195.0 (40.6)	.009	<.001
≥240 mg/dL, No. (%)					
Baseline	3838 (26.5)	2284 (26.6)	2178 (25.4)	.89	.06
2 Years	1898 (18.6)	1018 (16.9)	976 (17.0)	.005	.03
4 Years	1223 (14.4)	673 (13.4)	603 (12.8)	.13	.005
Potassium, mEq/L					
No. of participants (%)					
Baseline	14 487 (95.0)	8586 (94.9)	8573 (94.7)		
2 Years	9877 (64.7)	5794 (64.0)	5516 (60.9)		
4 Years	8315 (54.5)	4919 (54.4)	4616 (51.0)		
Mean (SD)					
Baseline	4.3 (0.7)	4.3 (0.7)	4.4 (0.7)	.59	.001
2 Years	4.0 (0.7)	4.3 (0.7)	4.5 (0.7)	<.001	<.001
4 Years	4.1 (0.7)	4.4 (0.7)	4.5 (0.7)	<.001	<.001
<3.5 mEq/L, No. (%)					
Baseline	493 (3.4)	292 (3.4)	223 (2.6)	.99	.001
2 Years	1254 (12.7)	151 (2.6)	83 (1.5)	<.001	<.001
4 Years	707 (8.5)	93 (1.9)	37 (0.8)	<.001	<.001
Fasting Glucose, mg/dL					
No. of participants (%)					
Baseline	11 273 (73.9)	6648 (73.5)	6752 (74.6)		
2 Years	5980 (39.2)	3506 (38.7)	3333 (36.8)		
4 Years	4972 (32.6)	2954 (32.6)	2731 (30.2)		
Mean (SD)					
Baseline	123.5 (58.3)	123.1 (57.0)	122.9 (56.1)	.71	.54
2 Years	127.6 (59.2)	122.4 (54.2)	120.8 (54.0)	<.001	<.001
4 Years	126.3 (55.6)	123.7 (52.0)	121.5 (51.3)	.20	.002
≥126 mg/dL, No. (%)					
Baseline	3258 (28.9)	1941 (29.2)	1985 (29.4)	.68	.55
2 Years	1967 (32.9)	1048 (29.9)	947 (28.4)	<.001	<.001
4 Years	1626 (32.7)	901 (30.5)	784 (28.7)	.11	<.001
Fasting Glucose Among Nondiabetics With Baseline Fasting Glucose <126 mg/dL					
No. of participants (%)					
Baseline	6766 (100)	3954 (100)	4096 (100)		
2 Years	3074 (45.4)	1787 (45.2)	1737 (42.4)		
4 Years	2606 (40.3)	1567 (39.6)	1464 (35.7)		
Mean (SD)					
Baseline	93.1 (11.7)	93.0 (11.4)	93.3 (11.8)	.52	.45
2 Years	102.2 (27.1)	99.0 (22.5)	97.4 (20.0)	<.001	<.001
4 Years	104.4 (28.5)	103.1 (27.7)	100.5 (19.5)	.11	<.001
≥126 mg/dL, No. (%)					
2 Years	295 (9.6)	132 (7.4)	101 (5.8)	.006	<.001
4 Years	302 (11.6)	154 (9.8)	119 (8.1)	.04	<.001

(continued)

amlodipine-based therapy in ALLHAT. In the Systolic Hypertension in the Elderly Program and the Systolic Hypertension in Europe trial, in which a thiazide-like diuretic (chlorthalidone) or a DHP-CCB was compared with a placebo, major CHD events were reduced by 27% and 30%, and stroke by 37% and 42%, respectively.[8,9] More direct evidence comes from 2 large active-controlled trials that compared DHP-CCB and traditional first-step drugs. The Swedish Trial in Old Patients with Hypertension-2 and the International Nifedipine GITS (long-acting gastrointestinal formulation) Study: Intervention as a Goal in Hypertension Treatment (INSIGHT), found no significant differences for major CHD or stroke rates between the treatment groups.[32,33] Some of these individual trials have had limited power to evaluate differences between treatments.[34] In meta-analyses of 5 positive-controlled trials, which included both DHP-CCB and non–DHP-CCB trials, there were trends that favored CCB-based therapy for stroke and traditional treatment for CHD, with no difference for all-cause mortality.[13,14] However, ALLHAT observed approximately the same number of strokes and nearly twice as many CHD events as all 5 trials combined, which suggests that the aggregate of the evidence would indicate no difference between CCB-based treatment and diuretic-based treatment for these outcomes.

The amlodipine vs chlorthalidone findings for HF reinforce previous trial results. In the diuretic-based Systolic Hypertension in the Elderly Program, active therapy reduced HF occurrence by 49% compared with placebo ($P<.001$), although in the DHP-CCB–based Systolic Hypertension in Europe trial, it was reduced by 29% (not statistically significant).[9,35] In the INSIGHT trial, HF was approximately twice as frequent in the CCB vs the diuretic arm.[33] The previously cited meta-analyses reported a higher rate of HF with CCB-based treatment than traditional regimens, with no difference in RR for DHPs compared with non–DHPs.[13,14]

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

A body of literature based on observational studies and secondary CHD prevention trials of short-acting CCBs has suggested that CCBs, especially DHP-CCBs, may increase the risk of cancer, gastrointestinal bleeding, and all-cause mortality.[14,36,37] The results of ALLHAT do not support these findings. In fact, the mortality from noncardiovascular causes was significantly lower in the CCB group (Table 6).

There were no significant differences in the incidence of ESRD between chlorthalidone and amlodipine, consistent with findings from the INSIGHT trial.[33] Comparison of the reciprocal serum creatinine slopes suggested a slower decline in kidney function in the amlodipine group. However, this finding requires cautious interpretation because studies assessing glomerular filtration rate more directly have shown a hemodynamically mediated acute increase in glomerular filtration rate followed by a more rapid rate of decline with chronic therapy using amlodipine and other CCBs.[38-40]

Comparison of the lisinopril and chlorthalidone groups revealed better drug tolerance and BP control with chlorthalidone. Angioedema, a rare but potentially serious adverse effect of ACE inhibitor use, occurred 4 times more frequently in participants randomized to lisinopril than in those randomized to chlorthalidone. Cholesterol levels, the prevalence of hypokalemia (serum potassium <3.5 mEq/L), and new diabetes (fasting glucose ≥126 mg/dL [≥7.0 mmol/L]) were higher in the chlorthalidone than the other groups following 2 and 4 years of follow-up. Overall, these metabolic differences did not translate into more cardiovascular events or into higher all-cause mortality in the chlorthalidone group compared with the other 2 groups.

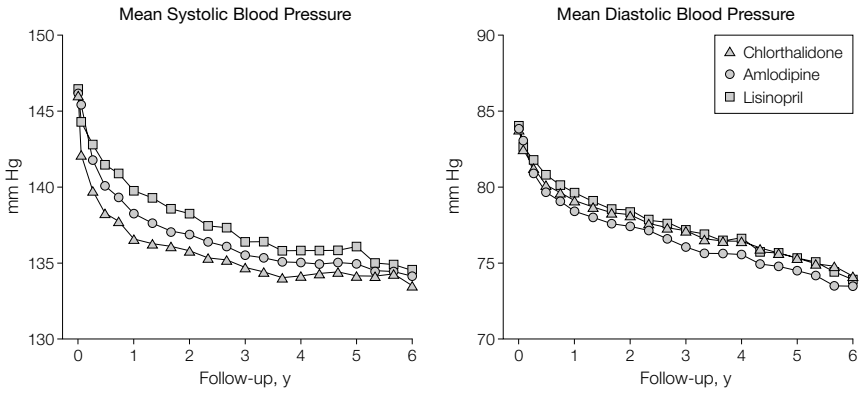
The ALLHAT findings for some major outcomes are consistent with predictions from placebo-controlled trials involving ACE inhibitors and diuretics. Specifically, for ACE inhibitor and diuretic trials, respectively, the reductions in CHD rates were 20% and 18%, and for all-cause mortality, 16% and 10%.[13] The 10% greater rate of combined CVD in the lisinopril than in the chlorthalidone group was due to increased occurrences of stroke, HF, angina, and coronary revascularization. Results for some of these outcomes may seem surprising, because of reports of beneficial effects of ACE inhibitors on surrogate markers of atherosclerosis and reductions in vascular and renal events in individuals with HF, diabetes, kidney disease, and cerebrovascular disease in placebo-controlled trials.[41-43] However, the finding in ALLHAT that HF incidence was lower in the diuretic vs the ACE inhibitor group is also consistent with previous reports. In the Systolic Hypertension in the Elderly Program trial (chlorthalidone vs placebo), there was a 49% decrease in the development of HF, whereas in the Studies of Left Ventricular Dysfunction Prevention (enalapril vs placebo) and Heart Outcomes Prevention Evaluation trials (ramipril vs placebo), there were only 20% and 23% reductions, respectively.[8,10,44] In published meta-analyses of placebo-controlled trials, the reductions in rates for stroke with ACE inhibitor and diuretics were 30% and 34%, translating into nearly equivalent results.[3,13] The 15% relative increase in stroke incidence for lisinopril compared with chlorthalidone treatment in ALLHAT must be considered in the context of heterogeneity of the results by race. The Swedish Trial in Old Patients with Hypertension-2 trial, which compared ACE inhibitors with conventional treatment (diuret-

Table 4. Biochemical Changes by Treatment Group* (cont)

	Chlorthalidone	Amlodipine	Lisinopril	*P* Value: Amlodipine vs Chlorthalidone	*P* Value: Lisinopril vs Chlorthalidone
Estimated Glomerular Filtration Rate, mL/min per 1.73 m²†					
No. of participants (%)					
Baseline	14 492 (95.0)	8589 (94.9)	8577 (94.7)		
2 Years	9877 (64.7)	5794 (64.0)	5516 (60.9)		
4 Years	8316 (54.5)	4924 (54.4)	4621 (51.0)		
Mean (SD)					
Baseline	77.6 (19.7)	78.0 (19.7)	77.7 (19.9)	.08	.57
2 Years	73.3 (19.9)	78.0 (20.5)	74.0 (20.0)	<.001	.002
4 Years	70.0 (19.7)	75.1 (20.7)	70.7 (20.1)	<.001	.03

*To convert serum cholesterol to mmol/L, multiply by 0.0259; fasting glucose to mmol/L, multiply by 0.0555.
†Simplified 4-variable Modification of Diet in Renal Disease Study formula.[24,25]

Figure 2. Mean Systolic and Diastolic Blood Pressure by Year During Follow-up



Number measured at baseline through 5 years is given in Table 3; numbers at 6 years for chlorthalidone, amlodipine, and lisinopril are 2721, 1656, and 1551, respectively.

©2002 American Medical Association. All rights reserved.

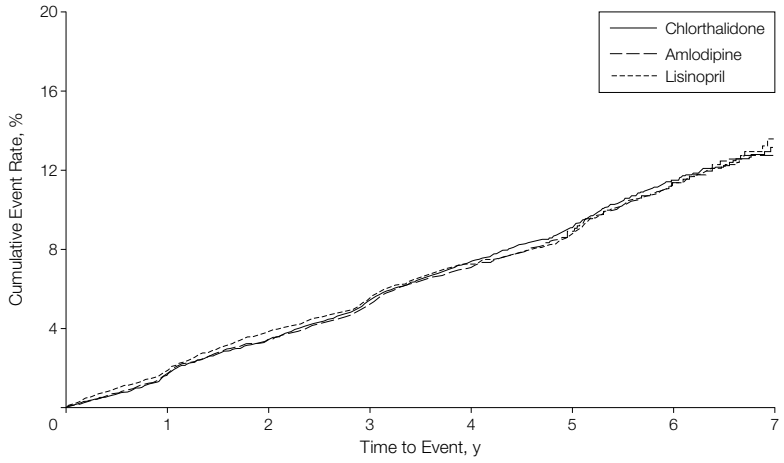
Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

Table 5. Clinical Outcomes by Antihypertensive Treatment Group*

	Chlorthalidone		Amlodipine		Lisinopril		Amlodipine vs Chlorthalidone			Lisinopril vs Chlorthalidone		
	No. of Total Events	6-Year Rate per 100 Persons (SE)	No. of Total Events	6-Year Rate per 100 Persons (SE)	No. of Total Events	6-Year Rate per 100 Persons (SE)	RR (95% CI)	Z Score	P Value	RR (95% CI)	Z Score	P Value
Primary outcome												
CHD†	1362	11.5 (0.3)	798	11.3 (0.4)	796	11.4 (0.4)	0.98 (0.90-1.07)	−0.46	.65	0.99 (0.91-1.08)	−0.24	.81
Secondary outcomes												
All-cause mortality	2203	17.3 (0.4)	1256	16.8 (0.5)	1314	17.2 (0.5)	0.96 (0.89-1.02)	−1.27	.20	1.00 (0.94-1.08)	0.12	.90
Combined CHD‡	2451	19.9 (0.4)	1466	19.9 (0.5)	1505	20.8 (0.5)	1.00 (0.94-1.07)	0.04	.97	1.05 (0.98-1.11)	1.35	.18
Stroke	675	5.6 (0.2)	377	5.4 (0.3)	457	6.3 (0.3)	0.93 (0.82-1.06)	−1.09	.28	1.15 (1.02-1.30)	2.31	.02
Combined CVD‡	3941	30.9 (0.5)	2432	32.0 (0.6)	2514	33.3 (0.6)	1.04 (0.99-1.09)	1.55	.12	1.10 (1.05-1.16)	3.78	<.001
End-stage renal disease	193	1.8 (0.1)	129	2.1 (0.2)	126	2.0 (0.2)	1.12 (0.89-1.40)	0.98	.33	1.11 (0.88-1.38)	0.87	.38
Cancer	1170	9.7 (0.3)	707	10.0 (0.4)	703	9.9 (0.4)	1.01 (0.92-1.11)	0.30	.77	1.02 (0.93-1.12)	0.42	.67
Hospitalized for gastrointestinal bleeding§	817	8.8 (0.3)	449	8.0 (0.4)	526	9.6 (0.4)	0.92 (0.82-1.03)	−1.44	.15	1.11 (0.99-1.24)	1.82	.07
Components of secondary outcomes												
Heart failure	870	7.7 (0.3)	706	10.2 (0.4)	612	8.7 (0.4)	1.38 (1.25-1.52)	6.29	<.001	1.19 (1.07-1.31)‖	3.33	<.001
Hospitalized/fatal heart failure	724	6.5 (0.3)	578	8.4 (0.4)	471	6.9 (0.4)	1.35 (1.21-1.50)	5.37	<.001	1.10 (0.98-1.23)‖	1.59	.11
Angina (hospitalized or treated)	1567	12.1 (0.3)	950	12.6 (0.4)	1019	13.6 (0.4)	1.02 (0.94-1.10)	0.42	.67	1.11 (1.03-1.20)	2.59	.01
Angina (hospitalized)	1078	8.6 (0.3)	630	8.4 (0.4)	693	9.6 (0.4)	0.98 (0.89-1.08)	−0.41	.68	1.09 (0.99-1.20)	1.85	.06
Coronary revascularizations	1113	9.2 (0.3)	725	10.0 (0.4)	718	10.2 (0.4)	1.09 (1.00-1.20)	1.88	.06	1.10 (1.00-1.21)	1.95	.05
Peripheral arterial disease (hospitalized or treated)	510	4.1 (0.2)	265	3.7 (0.2)	311	4.7 (0.4)	0.87 (0.75-1.01)	−1.86	.06	1.04 (0.90-1.19)	0.48	.63

*RR indicates relative risk; CI, confidence interval; CHD, coronary heart disease; and CVD, cardiovascular disease. CHD includes nonfatal myocardial infarction (MI) and fatal CHD; end-stage renal disease: kidney disease death, kidney transplant, or start of chronic renal dialysis; and heart failure: fatal, nonfatal hospitalized, or treated.
†Nonfatal MIs comprise 64% to 66% of the primary outcome.
‡Combined CHD indicates CHD death, nonfatal MI, coronary revascularization procedures, and hospitalized angina. Combined CVD indicates CHD death, nonfatal MI, stroke, coronary revascularization procedures, hospitalized or treated angina, treated or hospitalized heart failure, and peripheral arterial disease (hospitalized or outpatient revascularization).
§Denominators are 11 361 chlorthalidone, 6757 amlodipine, and 6665 lisinopril.
‖Proportional hazards assumption violated; data are RRs from a 2 × 2 table.

Figure 3. Cumulative Event Rates for the Primary Outcome (Fatal Coronary Heart Disease or Nonfatal Myocardial Infarction) by Treatment Group



No. at Risk								
Chlorthalidone	15 255	14 477	13 820	13 102	11 362	6340	2956	209
Amlodipine	9048	8576	8218	7843	6824	3870	1878	215
Lisinopril	9054	8535	8123	7711	6662	3832	1770	195

No significant difference was observed for amlodipine (relative risk [RR], 0.98; 95% confidence interval [CI], 0.90-1.07; $P = .65$) or lisinopril (RR, 0.99; 95% CI, 0.91-1.08; $P = .81$) vs chlorthalidone with a mean follow-up of 4.9 years.

ics and/or β-blockers), showed no significant differences in CHD, stroke, HF, or all-cause mortality.[32] Although these findings are somewhat different from the experience in ALLHAT, consideration needs to be given to respective confidence limits, population differences (especially race), and study designs (open vs double-blind).

No substantial differences in incidence of ESRD, glomerular filtration rate, or reciprocal creatinine slopes were noted for the lisinopril vs chlorthalidone comparisons. The ALLHAT study population was selected for high CVD risk and had a baseline mean creatinine of only 1.0 mg/dL (88.4 μmol/L). More detailed analyses of high renal risk subgroups (ie, diabetic, renal-impaired, and black patients) will be the subject of subsequent reports.

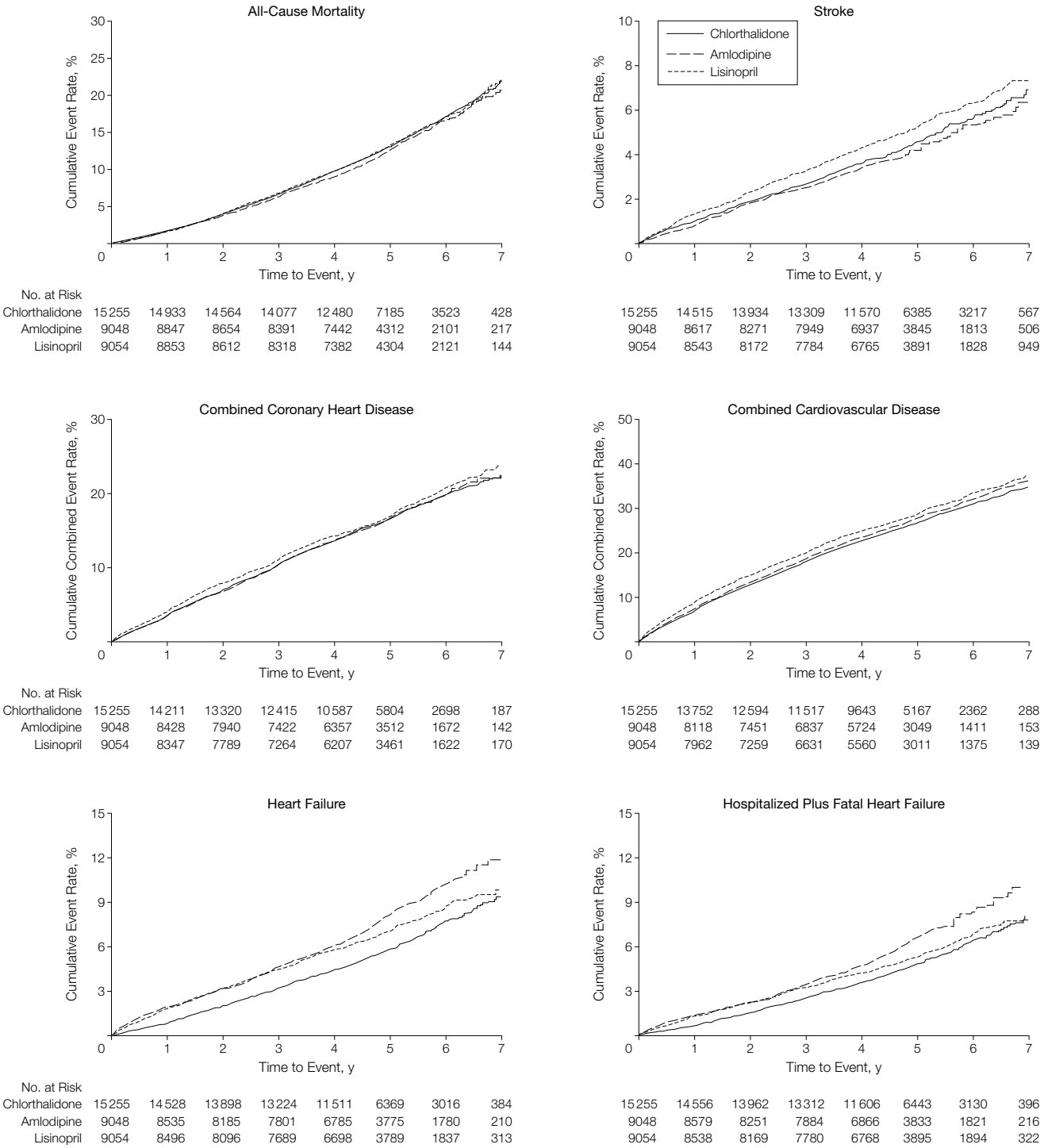
Analyses of RRs for stroke and HF adjusted for follow-up BP suggest that the 2-mm Hg systolic BP difference over-

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

all (4 mm Hg in black patients) between the lisinopril and chlorthalidone groups only partially accounts for the observed CVD event difference. However, such analyses are limited by the infrequency and imprecision of BP measurements for individual participants and regression dilution, which underestimates CVD risk associated

Figure 4. Cumulative Event Rates for All-Cause Mortality, Stroke, Combined Coronary Heart Disease, Combined Cardiovascular Disease, Heart Failure, and Hospitalized Plus Fatal Heart Failure by Treatment Group

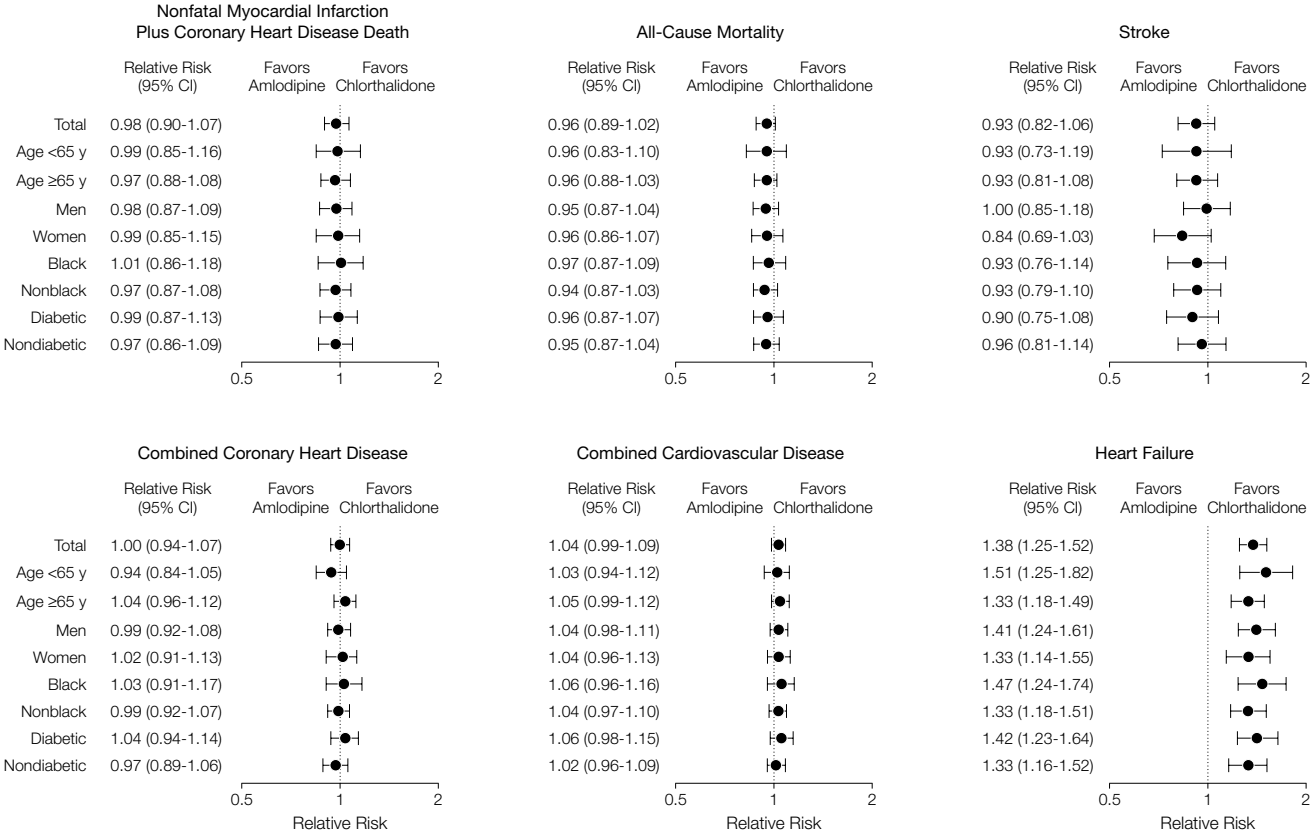


©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

with BP differences based on single-visit (or even visit-averaged) measurements.[45] Such modeling is also unable to account for differences among individuals due to other unmeasured or poorly represented risk factors; thus, participants who lower their BP by a given amount with one drug may not be comparable to those who lower their

Figure 5. Relative Risks and 95% Confidence Intervals (CIs) for Amlodipine/Chlorthalidone Comparisons in Prespecified Subgroups



Scales are shown in natural logarithm.

Table 6. Causes of Death by Antihypertensive Treatment Group*

	No. (6-Year Rate per 100 Persons)			*P* Value	
	Chlorthalidone (n = 15 255)	Amlodipine (n = 9048)	Lisinopril (n = 9054)	Amlodipine vs Chlorthalidone	Lisinopril vs Chlorthalidone
Total deaths	2187 (17.1)	1237 (16.5)	1303 (17.0)	.12	.90
Cardiovascular	992 (8.0)	592 (8.4)	609 (8.4)	.98	.53
Myocardial infarction	298 (2.4)	168 (2.3)	157 (2.2)	.56	.22
Definite CHD	118 (1.1)	74 (1.2)	78 (1.1)	.73	.47
Possible CHD	123 (1.1)	69 (1.1)	93 (1.4)	.68	.08
Stroke	163 (1.4)	91 (1.4)	116 (1.6)	.62	.14
Heart failure	116 (1.1)	79 (1.3)	68 (1.1)	.36	.92
Other CVD	174 (1.4)	111 (1.7)	97 (1.5)	.58	.62
Noncardiovascular	1058 (8.9)	559 (7.8)	606 (8.3)	.02	.47
Cancer	513 (4.3)	280 (3.7)	297 (4.0)	.23	.72
Kidney disease	36 (0.4)	23 (0.5)	28 (0.5)	.80	.29
Unintentional injury/suicide/homicide	65 (0.6)	18 (0.3)	27 (0.4)	.004	.12
Other non–CVD	444 (3.9)	238 (3.6)	254 (3.7)	.18	.62
Unknown	137 (1.2)	86 (1.2)	88 (1.3)	.72	.58

*CHD indicates coronary heart disease; CVD, cardiovascular disease.

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

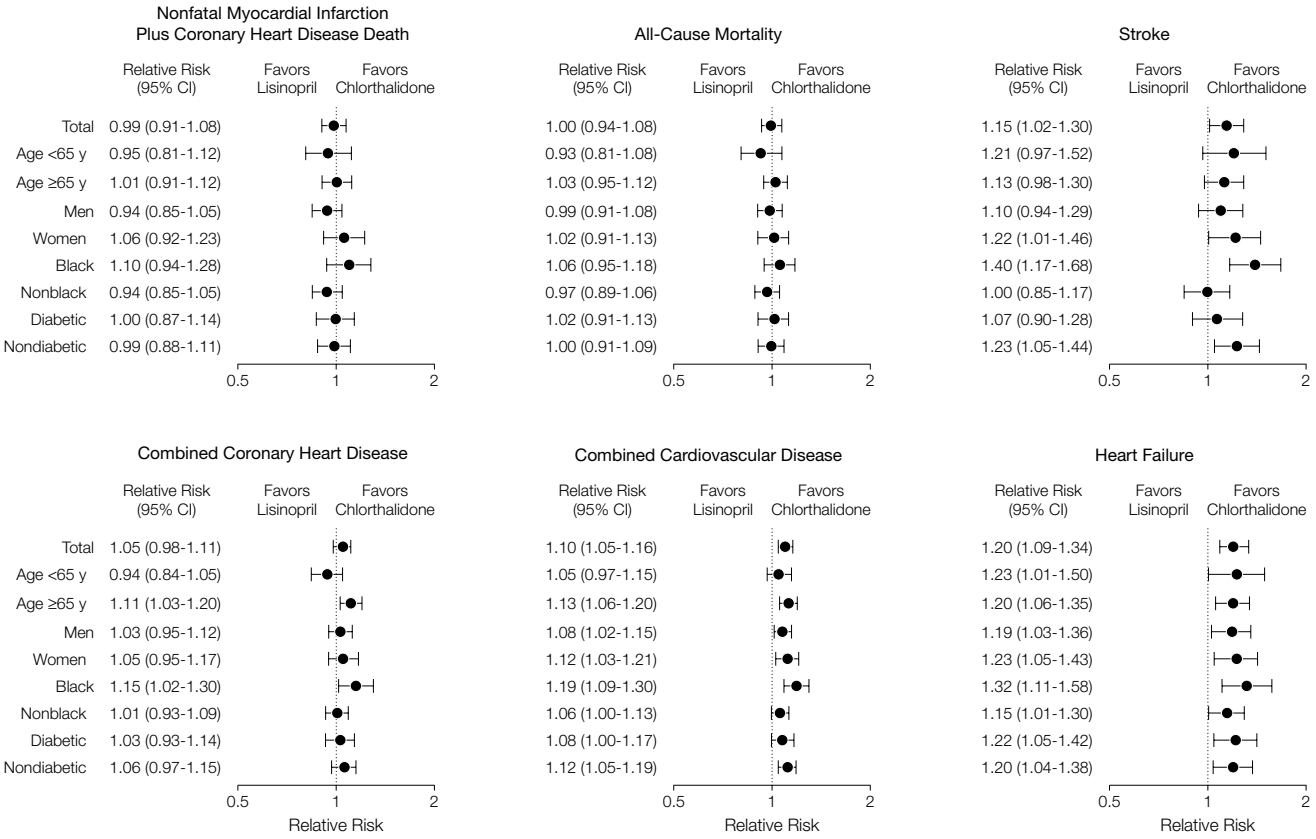
BP by the same magnitude with another drug.

Using an external standard of pooled results of long-term hypertension treatment trials and observational studies (10-12 mm Hg systolic BP difference associated with 38% stroke reduction), a 2- to 3-mm Hg difference in BP might account for a 6% to 12% difference in stroke rates.[45,46] This is consistent with the observed 15% difference for stroke overall but not with the difference seen in black patients (13%-16% expected, 40% observed). For the HF outcome, trial results in isolated systolic hypertension suggest that a 3-mm Hg higher systolic BP could explain a 10% to 20% increase in risk.[8,47] The forgoing ignores the absence of a diastolic BP difference in ALLHAT; however, the relationship of diastolic pressure and CVD events in elderly persons who often have increased pulse pressure is not entirely clear.[48]

The primary and secondary outcome results for the amlodipine vs chlorthalidone group comparisons were consistent for all subgroups of participants: older and younger, men and women, black and nonblack, diabetic and nondiabetic. For the lisinopril vs chlorthalidone comparisons, results were generally consistent by age, sex, and diabetic status. Thus, for the important diabetic population, lisinopril appeared to have no special advantage (and amlodipine no particular detrimental effect) for most CVD and renal outcomes when compared with chlorthalidone. In fact, chlorthalidone was superior to lisinopril for several CVD outcomes and superior to amlodipine for HF in both diabetic and nondiabetic participants. The consistency of the ALLHAT findings across multiple patient subgroups provides confidence in the ability to generalize the findings to most patients with hypertension.

In the lisinopril vs chlorthalidone comparisons, there were 2 outcomes with significant interactions. The greater differences observed in black vs nonblack patients for combined CVD and stroke, along with a similar trend for HF and lesser BP lowering with lisinopril, are in accord with the multiple reports of poorer BP response with ACE inhibitor in black patients.[49-51] They are also consistent with reports of lesser effects of ACE inhibitors in secondary prevention of HF in this population,[52,53] although these findings have been recently questioned.[54] The differential responses for disease outcomes parallel the lesser response in the black subgroup for BP, although the differences in outcomes are not substantially reduced by statistically adjusting for systolic BP.

Figure 6. Relative Risks and 95% Confidence Intervals (CIs) for Lisinopril/Chlorthalidone Comparisons in Prespecified Subgroups



Scales are shown in natural logarithm.

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

Although subordinate to safety and efficacy, the cost of drugs and medical care for the individual and society is a factor that should be considered in the selection of antihypertensives. One of the stated objectives of ALLHAT was to answer the question, "Are newer types of antihypertensive agents, which are currently more costly, as good or better than diuretics in reducing CHD incidence and progression?"[18] Consideration of drug cost could have a major impact on the nation's health care expenditures. Based on previous data that showed that diuretic use declined from 56% to 27% of antihypertensive prescriptions between 1982 and 1992, the health care system would have saved $3.1 billion in estimated cost of antihypertensive drugs had the pattern of prescriptions for treatment of hypertension remained at the 1982 level.[55] Further economic analyses based on the results of ALLHAT are under way.

The strengths of ALLHAT include its randomized double-blind design, statistical power to detect clinically meaningful differences in CVD outcomes of interest, diverse population with adequate representation from subgroups of special interest in the treatment of hypertension, and varied practice-based settings. In addition, the agents that were directly compared represent 3 of the most commonly used newer classes of antihypertensives vs the best studied of the older classes.

Some limitations are worth noting. After ALLHAT was designed, newer agents have been or may soon be released (eg, angiotensin-receptor blockers, selective aldosterone antagonists), which were not evaluated. Although clinical centers were blinded to the regimen and urged to achieve recommended BP goals, equivalent BP reduction was not fully achieved in the treatment groups. Furthermore, because diuretics, ACE inhibitors, CCBs, and α-blockers were evaluated in the trial, the agents available for step-up led to a somewhat artificial regimen (use of sympatholytics rather than diuretics and CCBs) of step-up drugs in the ACE inhibitor group. This may have contributed to the higher BPs in the ACE inhibitor group, especially in the black subgroup. However, mean follow-up BPs were well below 140/90 mm Hg in all treatment groups. Although ALLHAT did not compare a β-blocker to chlorthalidone, previous trials have suggested equivalence[45] or even inferiority[3] for major CVD events.

The ALLHAT results apply directly to chlorthalidone, amlodipine, and lisinopril. Combined with evidence from other trials, we infer that the findings also broadly apply to the drug classes (or subclass in the case of the dihydropyridine CCBs) that the study drugs represent. The evidence base for selection of antihypertensive agents has been markedly strengthened by the addition of ALLHAT.

In conclusion, the results of ALLHAT indicate that thiazide-type diuretics should be considered first for pharmacologic therapy in patients with hypertension. They are unsurpassed in lowering BP, reducing clinical events, and tolerability, and they are less costly. For patients who cannot take a diuretic (which should be an unusual circumstance), first-step therapy with CCBs and ACE inhibitors could be considered with due regard for their higher risk of 1 or more major manifestations of CVD. Since a large proportion of participants required more than 1 drug to control their BP, it is reasonable to infer that a diuretic be included in all multidrug regimens, if possible. Although diuretics already play a key role in most antihypertensive treatment recommendations, the findings of ALLHAT should be carefully evaluated by those responsible for clinical guidelines and be widely applied in patient care.

ALLHAT Authors/Officers and Coordinators: Curt D. Furberg, MD, PhD; Jackson T. Wright, Jr, MD, PhD; Barry R. Davis, MD, PhD; Jeffrey A. Cutler, MD, MPH; Michael Alderman, MD; Henry Black, MD; William Cushman, MD; Richard Grimm, MD, PhD; L. Julian Haywood, MD; Frans Leenen, MD; Suzanne Oparil, MD; Jeffrey Probstfield, MD; Paul Whelton, MD, MSc; Chuke Nwachuku, MA, MPH; David Gordon, MD, PhD; Michael Proschan, PhD; Paula Einhorn, MD, MS; Charles E. Ford, PhD; Linda B. Piller, MD, MPH; J. Kay Dunn, PhD; David Goff, MD, PhD; Sara Pressel, MS; Judy Bettencourt, MPH; Barbara deLeon, BA; Lara M. Simpson, MS; Joe Blanton, MS; Therese Geraci, MSN, RN, CS; Sandra M. Walsh, RN; Christine Nelson, RN, BSN; Mahboob Rahman, MD; Anne Juratovac, RN; Robert Pospisil, RN; Lillian Carroll, RN; Sheila Sullivan, BA; Jeanne Russo, BSN; Gail Barone, RN; Rudy Christian, MPH; Sharon Feldman, MPH; Tracy Lucente, MPH; David Calhoun, MD; Kim Jenkins, MPH; Peggy McDowell, RN; Janice Johnson, BS; Connie Kingry, RN, BSN; Juan Alzate, MD; Karen L. Margolis, MD; Leslie Ann Holland-Klemme, BA; Brenda Jaeger; Jeffrey Williamson, MD, MHS; Gail Louis, RN; Pamela Ragusa, RN, BSN; Angela Williard, RN, BSN; R. L. Sue Ferguson, RN; Joanna Tanner; John Eckfeldt, MD, PhD; Richard Crow, MD; John Pelosi, RPh, MS.

Financial Disclosures: The following listed authors have served as consultants for, received personal compensation from, were grant recipients of, or own stock in the following companies: *Furberg:* Merck, Pfizer, Pharmacia & Upjohn, Takeda, Wyeth-Ayerst; *Wright:* Aventis, Bayer, Bristol-Myers Squibb, Forrest Labs, King/Monarch, Merck, Novartis, Pfizer; *Davis:* Abbott, Bristol-Myers Squibb, Forrest Labs, Merck, Pfizer, Pharmacia & Upjohn, GlaxoSmithKline; *Alderman:* Bristol-Myers Squibb, Merck, Novartis, Pfizer, GlaxoSmithKline; *Black:* Abbott, AstraZeneca, Bristol-Myers Squibb, Novartis, Pfizer, Pharmacia & Upjohn, GlaxoSmithKline, Solvay; *Cushman:* AstraZeneca, Bristol-Myers Squibb, Forrest Labs, Merck, Pfizer, Pharmacia & Upjohn, Sankyo, Searle, Solvay, Takeda; *Grimm:* AstraZeneca, Merck, Novartis, Pfizer, Roche, Solvay; *Haywood:* Pharmacia & Upjohn; *Leenen:* AstraZeneca, Bayer, Bristol-Myers Squibb, Merck, Nu-Pharm, Pfizer, Pharmacia & Upjohn; *Oparil*: Abbott, AstraZeneca, Aventis, Bayer, Bristol-Myers Squibb, DuPont, Forrest Labs, King/Monarch, Merck, Novartis, Parke-Davis, Pfizer, Pharmacia & Upjohn, Roche, Sankyo, Schering-Plough, Searle, GlaxoSmithKline, Texas Biotechnology; *Probstfield:* AstraZeneca, King/Monarch, Pfizer; *Whelton:* Merck, Novartis, Pfizer, Pharmacia & Upjohn; *Ford:* Bristol-Myers Squibb; *Rahman:* Abbott, Novartis, Pfizer, Searle; *Lucente:* GlaxoSmithKline; *Calhoun:* AstraZeneca, Aventis, Merck, Novartis, GlaxoSmithKline; Solvay; *McDowell:* Amgen, King/Monarch, Merck, Pfizer; *Alzate:* Pfizer; *Tanner:* SGP; *Eckfeldt:* Johnson & Johnson.

Author Contributions: Dr Davis had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analyses in this article and its companion article on page 2998.

Study concept and design: Furberg, Wright, Davis, Cutler, Alderman, Black, Cushman, Grimm, Oparil, Whelton, Proschan, Ford, Piller, Goff, Lucente, Margolis, Williamson, Ragusa.

Acquisition of data: Wright, Davis, Alderman, Black, Cushman, Grimm, Haywood, Leenen, Oparil, Probstfield, Whelton, Einhorn, Ford, Piller, Pressel, deLeon, Simpson, Blanton, Geraci, Walsh, Nelson, Rahman, Juratovac, Pospisil, Carroll, Sullivan, Russo, Christian, Feldman, Lucente, Calhoun, Jenkins, McDowell, Johnson, Kingry, Alzate, Margolis, Holland, Jaeger, Williamson, Louis, Ragusa, Williard, Ferguson, Tanner, Eckfeldt, Crow, Pelosi.

Analysis and interpretation of data: Furberg, Wright, Davis, Cutler, Black, Cushman, Grimm, Haywood, Leenen, Oparil, Probstfield, Whelton, Nwachuku, Gordon, Proschan, Einhorn, Ford, Piller, Dunn, Goff, Pressel, Bettencourt, Simpson, Rahman, Barone, Williamson.

Drafting of the manuscript: Furberg, Wright, Davis, Cutler, Alderman, Black, Cushman, Grimm, Haywood, Leenan, Oparil, Probstfield, Whelton, Nwachuku, Gordon, Proschan, Einhorn, Ford, Piller, Dunn, Goff, Pressel, Bettencourt, Simpson, Rahman, Kingry, Margolis, Williamson.

Critical revision of the manuscript for important intellectual content: Furberg, Wright, Davis, Cutler, Alderman, Black, Grimm, Haywood, Leenen, Oparil, Probstfield, Whelton, Nwachuku, Gordon, Proschan, Einhorn, Ford, Piller, Dunn, Goff, Pressel, Bettencourt,

 ©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

deLeon, Simpson, Geraci, Walsh, Rahman, Pospisil, Carroll, Sullivan, Russo, Barone, Christian, Feldman, Lucente, Calhoun, Jenkins, McDowell, Johnson, Kingry, Alzate, Margolis, Williamson, Louis, Williard, Ferguson, Tanner, Pelosi.

Statistical expertise: Davis, Whelton, Proschan, Ford, Dunn, Pressel.

Obtained funding: Davis, Cutler, Black, Einhorn, Ford, Goff, Sullivan.

Administrative, technical, or material support: Furberg, Wright, Davis, Cutler, Alderman, Black, Cushman, Grimm, Haywood, Oparil, Probstfield, Whelton, Nwachuku, Gordon, Einhorn, Ford, Piller, Pressel, Bettencourt, deLeon, Simpson, Blanton, Geraci, Walsh, Nelson, Rahman, Juratovac, Pospisil, Carroll, Russo, Barone, Christian, Feldman, Lucente, Jenkins, McDowell, Johnson, Kingry, Alzate, Margolis, Holland, Jaeger, Louis, Williard, Ferguson, Tanner, Eckfeldt, Pelosi.

Study supervision: Furberg, Wright, Davis, Cutler, Black, Cushman, Grimm, Haywood, Leenen, Oparil, Probstfield, Ford, Pressel, Lucente, Alzate, Holland, Jaeger, Eckfeldt.

Funding/Support: This study was supported by contract NO1-HC-35130 with the National Heart, Lung, and Blood Institute (NHLBI). ALLHAT investigators received contributions of study medications supplied by Pfizer (amlodipine and doxazosin), AstraZeneca (atenolol and lisinopril), and Bristol-Myers Squibb (pravastatin), and financial support provided by Pfizer.

Role of the Sponsor: The NHLBI sponsored the study and was involved in all aspects other than direct operations of the study centers. This included collection, analysis, and interpretation of the data in addition to the decision to submit the manuscript for publication.

Dedication: Special recognition is due to 3 ALLHAT leaders who died in recent years after making very significant contributions to initiating the trial and overseeing most of its course: Richard Carleton, MD, Chairman of the Data and Safety Monitoring Board (1994-2000); H. Mitchell Perry, Jr, MD, member of the Steering Committee and Deputy Physician Coordinator for Region 1 (1994-2001); and Peter Frommer, MD, National Heart, Lung, and Blood Institute Deputy Director Emeritus, advisor to the Project Officers, and liaison to participating pharmaceutical companies (1993-2002).

Members of the ALLHAT Group: *Steering Committee:* Furberg, Wright, Davis, Cutler, Alderman, Black, Cushman, Grimm, Haywood, Leenen, Oparil, Probstfield, Whelton; *NHLBI Project Office:* Cutler, Nwachuku, Gordon, Proschan, Einhorn; *ALLHAT Clinical Trials Center:* Davis, Ford, Piller, Dunn, Pressel, Bettencourt, deLeon, Simpson, Blanton; *ALLHAT Regions: Veterans Administration, Memphis, Tenn:* Cushman, Geraci, Walsh, Nelson; *Cleveland, Ohio:* Wright, Rahman, Juratovac, Pospisil, Suhan; *Bronx, NY:* Alderman, Carroll, Russo, Sullivan; *Chicago, Ill:* Black, Barone, Christian, Feldman, Lucente; *Birmingham, Ala:* Oparil, Calhoun, Jenkins, McDowell; *Seattle, Wash:* Probstfield; Alzate, Johnson, Kingry; *Minneapolis, Minn:* Grimm, Margolis, Holland, Jaeger; *New Orleans, La (formerly located in Baltimore, Md):* Whelton, Williamson, Louis, Ragusa, Williard, Adler; *Ottawa, Ontario, Canada:* Leenen, Ferguson, Tanner; *ALLHAT Central Laboratory:* J. Eckfeldt, J. Bucksa, M. Nowicki; *ALLHAT Drug Distribution Center:* J. Pelosi; *ALLHAT Electrocardiogram Reading Center:* R. Crow, S. Thomas; *ALLHAT Data and Safety Monitoring Board:* R. Califf, W. Applegate, J. Buring, E. Cooper, K. Ferdinand, M. Fisher, R. Gifford, S. Sheps.

Investigators and Coordinators Participating in the Antihypertensive and Lipid Trials, United States: Alabama: L. Ada, D. Alexander, L. Black, C. Davis, W. Davis, S. Farooqui, H. Fritz, T. Kessler, S. Ledbetter, L. Means, J. Patterson, N. Qureshi, L. Redcross, R. Reeves, T. Tucker, N. Wettermark, A. Williams, W. Yarbrough; *Arizona:* I. Cohen, W. Dachman, N. Estrada, J. Felicetta, D. Fowler, R. Fowler, S. Goldman, C. Lui, S. Morris, D. Morrison, J. Nelson, J. Ohm, D. Paull, G. Pulliam, D. Roberts, I. Ruiz, H. Thai; *Arkansas:* J. Acklin, M. Azhar, F. Berry, D. Burns, W. Carter, M. Dixon, S. Eldridge, A. Fendley, H. Fendley, M. Flowers, S. Goss, M. Guyer, G. Harris, M. Hawkins, D. Hopson, P. Kern, R. King, M. Lynch, E. Maples, R. McCafferty, M. McGehee, J. Miller, D. Neil, M. Oakum, N. Paslidis, K. Riordan, G. Robbins, D. Simmons, C. Vilayvanh, S. Whitmer; *California:* C. Alvarez, D. Anderson, M. Ariani, S. Barrett, J. Boggess, B. Brackeen, A. Bui, P. Callaham, M. Calong, J. Camacho, J. Cavendish, G. Chao, D. Cheung, B. Christianson, W. Dempsey, G. Dennish, V. DeQuattro, R. Dharawat, D. Dizmang, N. Doherty, M. Donnell, S. Edmondson, D. Falcone, S. Franklin, J. Frazee, G. Frivold, S. Ghattas, D. Goldfarb-Waysman, T. Haskett, L. Haywood, N. Horton, Y. Huang, K. Hui, N. Jacob, K. Jolley, B. Jurado, A. Karns, R. Karns, K. Karunaratne, A. Katchem, L. Katchem, J. Khoo, E. Kiger, L. Kleinman, J. Kozlowski, D. Kramer, E. Lee, D. Li, C. Libanati, P. Linz, D. Lyle, T. Maekawa, M. Mahig, J. Mallery, D. Martins, B. Massie, R. Mikelionis, S. Myers, J. Neutel, N. Nguyen, U. Okoronkwo, K. Owens, T. Pan, R. Petersen, A. Schultz, H. Schultz, E. Schwartz, J. Schwartz, P. Schwartz, C. Scott, Z. Song, J. Taylor, D. Townsend, S. Turitzin, D. Ujiiye, A. Usman, D. Van Ostaeyen, R. Wadlington, C. Wan, L. Wang, H. Ward, L. Wieland, P. Williams-Brown, N. Wong, R. Wright; *Colorado:* K. Castleman, M. Chase, R. Hildenbrand, P. Lowe, P. Mehler, S. Mroz, R. Simpson, R. Tello; *Connecticut*: J. Bernene, L. Ciarcia, A. Grover, J. Judge, A. Lachman, J. Lawson, N. Medina, E. Nestler, R. Schwartz, B. Sicignano, S. Solinsky; *Washington, DC:* J. Golden, E. Lewis, D. Mateski, P. Narayan, A. Notargiacomo, D. Ordor, V. Papademetriou, O. Randall, T. Retta, J. Theobalds, S. Xu; *Delaware:* D. Crane, J. Lenhard; *Florida:* K. Anderson, S. Beery, G. Bhaskar, B. Booker, K. Broderick, E. Capili-Rosenkranz, J. Ciocon, G. Cohn, T. Connelly, V. Dallas, G. Duren, J. Durr, J. Evans, S. Feld, R. Feldman, L. Fischer, S. Fisher, M. Formoso, S. Fulford, M. Galler, J. Hildner, K. Holman, A. Jackson, C. Jackson, G. Khan, M. Khan, S. Kronen, J. Lehmann, A. Littles, R. Lopez, N. Madhany, L. McCarty, K. Mullinax, M. Murray, J. Navas, A. Peguero-Rivera, R. Preston, N. Rolbiecki, J. Rolle, L. Rosenfield, O. Saavedra, A. Schlau, M. Stein, J. Stokes, S. Strickland, U. Tran, B. Videau, J. Webster, T. Webster, A. Weinstein, T. Westfall, D. Williams, M. Yoham; *Georgia:* D. Anderson, R. Anderson, J. Barzilay, S. Boyce, P. Brackett, P. Bradley, W. Brown, R. Carter, S. Carter, D. Castro, L. Duty, H. Ellison, A. Francis, L. Goodman, D. Harrelson, T. Hartney, J. Heldreth, J. Heneisen, A. Hicks, L. Hornsby, J. Hudson, S. Hurst, L. Iskhakova; S. James, S. James, Y. Jones, K. Kersey, W. Kitchens, N. London, M. Loraditch, G. Lowe, R. Maddox, R. Malcolm, D. Mathis, C. Mayers, M. McDaniel, N. McPhail, A. Mikhail, H. Muecke, R. Noel, W. North, N. Parikh, D. Parish, G. Peters, P. Poulos, M. Ram, W. Rawlings, R. Remler, C. Rice, M. Salles, D. Sauers, A. Scheetz, C. Scott, L. Stevenson, J. Sumner, M. Sweeney, E. Taylor, K. Upadhya, T. Vu, M. Walsh, K. Williams, H. Yager; *Illinois:* M. Arron, C. Bareis, J. Barnett, G. Barone, C. Bermele, T. Bertucci, J. Cheng, J. Cruz, T. Denecke-Dattalo, S. Durfee, E. Edwards, L. Fahrner, D. Farley, T. Flegel, M. Friedman, C. Gaca, J. Gilden, S. Goldman, J. Graumlich, A. Hoffman, K. Hunt, C. Johnson, P. Kellums, A. Lasala, N. Lasala, V. Lauderdale, M. Lesko, F. Lopez, M. Mansuri, S. Mansuri, M. Martin, L. Moody, L. Morowczyneski, S. Mouritzen, N. Novotny, A. Ovalle, P. Pedersen, N. Perlman, P. Porcelli, B. Ragona, R. Sadiq, P. Sands, C. Simmons, K. Stevens, G. Sussman, D. Vicencio, A. Villafria, R. Villafria, R. Watkins; *Indiana:* J. Addo, J. Beliles, V. Dave, D. Fausset, J. Fox, D. Fryman, J. Hall, J. Koehler, L. Leavy, P. Linden, E. Long, H. Macabalitaw, T. Nguyen, B. Peterson, J. Pratt, D. Rosanwo, D. Ross, H. Shah, V. Shah, T. Smith, M. Sobol, B. Viellieu-Fischer, J. Wachs, B. Weinberg; *Iowa:* V. Butler, A. Durbin, R. Glynn, B. Hargens, W. Lawton, M. Roberts, J. Roepke, R. Schneider, G. Stanley; *Idaho:* M. Baker, R. Force, T. Gillespie, S. Hillman, K. Krell, M. Macdonald; *Kansas:* D. Courtney, B. Crawford, D. DeVore, J. Moppin, N. Premsingh, K. Reuben-Hallock, R. Schanker, D. Wilson; *Kentucky:* R. Berkley, M. DeMuro, L. Kazmierzak, A. Rayner, C. Tyler, E. Wells, S. Winters; *Louisiana:* E. Aguilar, L. Bass, V. Batuman, B. Beard, L. Borrouso, M. Campbell, C. Chubb, P. Connor, C. Conravey, D. Doucet, M. Doucet, J. Dunnick, D. Eldridge, T. Eldridge, P. Galvan, A. Gupta, J. Hollman, D. Hull, B. Jackson, T. Jones, A. Klenk, P. Lakshmiprasad, B. Mahl, J. Paranilam, E. Reisin, H. Rothschild, J. Sampson, B. Samuels, J. Schmitt, A. Smith, V. Valentino, C. Verrett, P. Willhoit; *Maine:* B. Blake, T. Lebrun, C. Walworth, R. Weiss; *Maryland*: J. Burton, W. Carr, P. Chance, S. Childs, C. Compton, J. Cook, V. Coombs, J. Daniels, P. Death, L. Essandoh, Y. Ferguson, D. Fraley, M. Freedman, M. Gary, F. Gloth, S. Gottlieb, M. Gregory, S. Hairston, P. Hall, B. Hamilton, J. Hamilton, D. Harrison, D. James, B. Kerzner, A. Lancaster, H. Lutz, J. Marks, J. Martin, J. Mersey, L. Nelson, E. Obah, S. Ong, J. Palacios, S. Park, M. Partlow, M. Posner, H. Rachocka, M. Rubin, M. Rubinstein, M. Rykiel, C. Smith, B. Socha, K. Thompson, K. Walker, J. Webber, K. Williams; *Massachusetts:* L. Bradshaw, A. Chakraborty, F. DiMario, J. Ingelfinger, J. Pincus, A. Sobrado; *Michigan:* L. Bey-Knight, D. Carson, A. Cavanaugh, M. Chertok, K. Church, H. Colfer, I. Diaz, B. Dobbs, G. Edelson, J. Fabello-Gamiao, S. Gappy, J. Grove, D. Johnson, M. Johnson, C. Jones, E. Jones, T. Kelly, N. Kerin, B. Letzring, M. Oleszkowicz, A. Raffee, K. Rasikas, C. Shaw, M. Siddique, B. VanOver, M. Zervos; *Minnesota:* D. Berman, V. Canzanello, J. Curtis, V. Erickson, W. Goodall, J. Graves, K. Guthrie, J. Haight, S. Hassing, J. Heegard, J. Holtzman, D. Jespersen, L. Klein, C. Kubajak, L. Nylund, P. Spilseth; *Missouri:* B. Appleton, R. Baird, S. Carmody, C. Carter, F. Charles, T. Finnigan, S. Giddings, K. Gorman, M. Gregory, L. Johnson, S. Joseph, L. Kennington, R. Kevorkian, J. LaSalle, B. Nolfo, J. Nunnelee, A. Orf, D. Palmer, H. Perry, A. Quick, B. Rogers, B. Rosemergey, C. Scott, S. Sharma, V. Shortino, D. Smith, K. Smith, C. Stanford, C. Tudor, T. Wiegmann; *Mississippi:* C. Adair, S. Armstrong, C. Brown, N. Brown, R. Brown, S. Burke, L. Burrell, L. Clark, S. Cooks, W. Crowell, D. Ellis, D. Graham, V. Green, R. Hall, S. Hamler, D. Haymon, A. Hinton, M. Holman, A. James, P. Karim, K. Kirchner, A. Knotts, A. Lott, W. McArthur, F. McCune, B. Miller, H. Morrow, R. Murphy, R. Myers, S. Myers, A. Phillips, M. Puckett, E. Rankin, O. Ransome-Kuti, M. Reddix, R. Rigsby, E. Searcy, D. Smith, A. Spann, Y. Tanner, E. Taylor-McCune, J. Tramuta, H. Wheeler, M. Wofford; *Montana:* L. Bigwood-Pecarina, S. English, H. Knapp, L. Sokoloski; *Nebraska:* M. Berry, E. Butkus, S. Byers, D. Colan, R. Dobesh, N. Hilleman, R. Hranac, P. Klein, T. McKnight, S. Mohiuddin, A. Mooss, R. Moyer, P. Myers, L. Rasmussen, J. Schafersman; *Nevada:* J. Chinn, R. Collins, E. Samols; *New Jersey:* S. Akgun, A. Bastian, L. Bordone, N. Cosgrove, A. Costa, A. Cuyjet, S. Daniels, L. DeEugenio, L. DeEugenio, R. Denniston, L. Duh, M. Farber, M. Farber, S. Ferguson, K. Ferranti, G. Flanagan, J. Garofalo, H. Hassman, J. Hassman, H. Jacobs, J. Kostis, A. Kudryk, M. Kutza, R. Liang, G. McArthur, B. McGann, R. Miller, E. Moser, F. Nash, P. Niblack, E. Ogunmefun, M. Raghuwanshi, S. Sastrasinh, T. Seely, J. Stanley, S. Suarez, A. Vaughn, R. Wong-Liang, J. Young, S. Yuchnovitz, M. Zolnowski; *New Mexico:* D. Graves, M. Groves, E. Iwan, J. Shipley; *New York:* N. Almelda, S. Anderson, J. Andres, N. Ankomah, E. Anteola, C. Assadi, M. Assadi, S. Atlas, J. Baruth, D. Barz, J. Begley, T. Bharathan, A. Bova, D. Brautigam, C. Brown, S. Canaan, M. Candelas, P. Caraballo, J. Chapman, L. Clark, K. Desai, D. Dowie, C. Dwyer, A. Farag, C.

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

Flanders, P. Foster, L. Gage, A. Gartung, S. Gedan, P. Gehring, J. Gorkin, D. Graber, H. Guber, P. Gugliuzza, J. Halbach, A. Henriquez, M. Henriquez, D. Hoffman, J. Holland, C. Hopkins, C. Hull, E. Ilamathi, K. Johnston, M. Karim, L. Katz, K. Kellick, S. Kerlen, M. Krishnamurthy, D. Lainoff, R. Levin, V. Littauer, J. Lohr, M. Lorenz, C. Lynott, J. Maddi, L. Marquart, K. Martin, M. Maw, R. Mendelson, S. Monrad, A. Mustapha, A. Nafziger, M. Neary, J. Ngheim, A. Niarchos, M. Noor, M. Omoh, J. Pickard, M. Pier, V. Pogue, C. Reddy, J. Ringstad, T. Rocco, C. Rosendorff, H. Sandefur, A. Sass, R. Schifeling, D. Scott, P. Scriber, K. Sharma, C. Shmukler, D. Shrivastava, M. Siegelheim, G. Smith, B. Snyder, C. Spiller, M. Srivastava, S. Stevenson, A. Stewart, B. Sumner, M. Sweeney, K. Thomas, L. Thomas, L. Trawlick, N. Velez, J. Vento, H. Viswaswariah, M. Yevdayeva, D. Zimmerman; *North Carolina:* T. Barringer, V. Bland, M. Burke-Ziglar, K. Caldwell, R. Caldwell, F. Celestino, G. Cole, M. Darrow, B. Dunn, S. Fox, J. Holbrook, K. Jacobs, J. Lisane, L. Loggans, A. Lowdermilk, R. Merrill, P. Miller, C. Perkins, L. Rodebaugh, V. Schlau, R. Smith, J. Spruill, J. Summerson; *North Dakota:* N. Chelliah, E. Garten, K. Hagen, S. Jafri, D. Vold, B. Westacott; *Ohio:* L. Barnes-Lark, C. Blanck, K. Casterline, D. Chen, K. Cowens, M. Cubick, D. Davidson, P. Dockery, J. Finocchio, T. Gundrum, T. Hentenaar, D. Hulisz, D. Hull, K. Keaton, G. Kikano, K. Klyn, L. Lazaron, D. Lukie, S. Medwid, L. Miller, R. Murden, H. Neff, E. Ospelt, M. Patel, E. Pelecanos, E. Pfister, L. Sadler, M. Saklayen, A. Salomon, A. Schmidt, S. Stein, D. Subich, D. Thiel, L. Thompson, R. Toltzis, J. Tucker, D. Vidt, G. Wise, D. Wray; *Oklahoma:* D. Abott, J. Cook-Greenwood, M. Jelley, R. Kipperman, J. Leverett, C. Manion, S. Mears, B. Parker, R. Ringrose, L. Scholl, J. Schoshke, F. Shelton, M. Stephens, U. Thadani, K. Walters; *Oregon:* M. Dissanayake, S. Falley, H. Harris, S. MacKenzie, F. McBarron, S. Murray; *Pennsylvania:* G. Abbott, C. Baessler, M. Benioff, A. Bowens, J. Burke, L. Carradine, K. Devine, M. Duzy, G. Dy, J. Fontaine, D. Fox, W. Gilhool, J. Grasso, T. Ham, S. Heaney, J. Hefner, D. Herr, L. Hollywood, L. Jones, M. Kauffman, E. Kemler, S. Koduri, N. Kopyt, S. Kutalek, M. MacIntyre, R. Martsolf, A. McLeod, A. Miller, A. Minnock, Y. Mishriki, D. Nace, L. Nagy, R. Olasin, C. Oschwald, N. Potts, R. Reinhard, R. Reinhard, N. Roberts, B. Rogers, D. Sant Ram, F. Sessoms, M. Shore, S. Shore, D. Singley, J. Spencer, D. Spigner, B. Springer, W. Swagler, P. Tanzer, S. Walker, N. Walls, D. Whyte, S. Worley, G. Ziady; *Puerto Rico:* A. Agosto, J. Aguilera-Montalvo, H. Algarin-Sanchez, J. Alvarado, I. Andino, J. Aponte Pagan, M. Arce, J. Benabe, J. Cangiano, L. Catoni, J. Cianchini, J. Claudio, M. Collazo, P. Colon, Y. Cruz-Lugo, J. DaMore, E. Edwards Volquez, A. Feliberti-Irizarri, P. Felix-Ramos, J. Fernandez-Quintero, M. Geo, M. Gomez, R. Gomez Adrover, L. Gonzalez-Bermudez, M. Guerrero, E. Guzman, J. Heredia, C. Irizarry, A. Leon, T. Lugardo, G. Martinez, R. Martinez, M. Melendez, M. Natal, M. Padilla, W. Pagan, Z. Perez, J. Pimentel, M. Pimentel Lebron, A. Ramos, M. Rios, C. Rivera, E. Rivera, J. Rivera Santiago, E. Rodriquez, D. Romero, R. Ruiz, C. Sanchez, J. Sanchez, M. Sosa-Padilla, I. Sotomayor-Gonzalez, J. Tavarez, I. Toro-Grajales, B. Torres, N. Vazquez, S. Vazquez, M. Vega, Z. Vidal Oviedo, V. Zapata, I. Zayas-Toro; *Rhode Island*: C. Alteri, J. Galli, A. Hordes, L. Laflamme, K. MacLean, L. Marquis, R. Ruggieri, S. Sharma; *South Carolina:* J. Basile, L. Clarke, I. Coley, D. Devlin, S. Eggleston, G. Goforth, D. Ham, A. Hampton, P. Hill, K. Jones, R. Jones, P. Jumper, A. Kitchens, C. Lieberman, J. McAlpine, J. Moloo, A. Saenz, D. Sheek, A. Smith-Salley, P. Snape, J. Sterrett, C. Stone, M. Strossner, C. Sullivan, T. Vear, D. Weathers, M. Weeks, J. Williams, M. Williams; *South Dakota:* C. Ageton, M. Brown, L. Dale, L. Duncan, S. Eckrich, P. Kearns, B. Lankhorst, K. McDougall, V. Schuster, J. Wegenke, J. Woehl, E. Zawada; *Tennessee:* D. Anderson, C. Bounds, J. Caldwell, W. Cannon, R. Cassidy, W. Cushman, C. DeJesus, L. Dilworth, S. Duffy, B. Hamilton, T. Harrell, K. Harris, M. Herr, J. Jones, L. Jones, H. Marker, J. Miller, S. Miller, F. Putman, A. Reaves, V. Rhule, H. Ross-Clunis, S. Satterfield, G. Siami, R. Smith, A. Smuckler, C. Snorton, T. Stern, D. Venugopal; *Texas:* A. Abbas, H. Adrogue, A. Amador, L. Arango, C. Arroyo, V. Battles, M. Beard, J. Beasley, R. Bhalla, G. Chauca, P. Damico, S. Davison, P. Dlabal, N. Duronio, C. East, F. Eelani, C. Farmerie, E. Fowler, O. Gambini, E. Griego, G. Habib, S. Hanna, D. Harden, T. Harrington, C. Herrera, T. Hicks, B. Hiltscher, D. Hyman, I. Lalani, A. Levine, S. Lu, I. Martinez, Y. Martinez, N. Mata, R. Motaparthi, B. Norch, M. Ottosen, V. Pavlik, L. Pearce, J. Periman, M. Pickard, N. Pokala, A. Ray, D. Richard, K. Rogers, M. Ruggles, L. Seals, D. Shafer, T. Shamsi, D. Sherwood-Berner, E. Soltero, A. Sy, J. Tomlinson, C. Vallbona, D. Verrett, R. Victor, W. Vongpatanasin, R. Young; *Utah:* R. Callihan, G. Henderson, J. O'Donnell, C. Slot, J. Swauger, C. Westenfelder, C. Williams; *Vermont:* B. Armstrong, B. Buckley, P. Courchesne, P. Cushman, F. Gallant, T. Howard, J. Osborne, R. Primeau, T. Tanner; *Virgin Islands:* K. Bryan-Christian, C. Christian, M. Morris; *Virginia:* D. Bryan, D. Connito, K. Damico, L. Gendron, E. Goudreau, M. Juarez, R. Lemly, L. Macklin, K. McCall, J. Moore, D. Panebianco, D. Paulson, A. Pemberton, R. Renzi, D. Rice, J. Schmitt, S. Speese, J. Sperling, L. Thompson, G. Vetrovec, A. Williams, D. Williams, B. Zambrana; *Washington:* J. Anderson, K. Capoccia, G. Deger, A. Ellsworth, A. Micketti, W. Neighbor, S. Yarnall; *West Virginia:* H. Blackwood, S. Grubb; *Wisconsin:* P. Ackell, A. Arnold, S. Blumenthal, P. Bodmer, R. Dart, D. David, D. Duffy, L. Egbujiobi, M. Faignant, A. Friedman, B. Friedman, C. Koeppl, M. Lintereur, J. Morledge, D. Neu, M. Noble, M. Rassier, G. Shove, M. Stevens, R. Wergin, L. Wollet, B. Yug, C. Zyniecki; *Investigators and Coordinators, Canada: New Brunswick:* C. Baer, J. LeBlanc, R. Withers, J. Yang; *Newfoundland:* J. Collingwood, P. Crocker, F. Jardine, S. Newman, G. Rideout, B. Sussex; *Ontario:* J. Baker, D. Bishop, C. Brose, D. Carswell, L. Charles, D. Coates, E. Coletta, M. Courtland, S. Crocker, R. Dhaliwal, T. Doey, D. Guy, D. Harterre, G. Harterre, C. Henry, D. Henry, D. Hutton, I. Janzen, H. Kafka, W. Kendrick, N. Kumar, R. Lan, F. Leenen, R. Lovell, B. McAuley, B. Melbourne, S. Melbourne, H. Morwood, S. Munro, S. Nawaz, T. O'Callahan, S. Prasad, P. Richardson, R. Rose, C. Sanderson-Guy, N. Schmidt, D. Spink, P. Spink, A. Stajfer, R. Tee, K. Usher, M. Wahby, R. Wahby, D. Wattam, L. Wells, M. Wiebe, K. Zarnke, P. Zuliani; *Prince Edward Island:* D. Cameron.

Investigators and Coordinators Participating in the Antihypertensive Trial Only, United States: California: P. Bailey-Walton, N. Bednarski, M. Chen, S. Fochler, S. Gross, T. Harper, G. Hilliard, B. Holmes, E. Jacobson, P. Kirkland, N. Lepor, K. Moorehead, E. Portnoy, S. Rieux, N. Rodriguez, D. Schneidman, F. Yuen; *Delaware:* J. Holleger, T. Tonwe; *Florida:* U. Anderson, B. Austin, L. Bianco, F. Griffith, J. Jaffe, E. Killeavy, A. Kwon, C. Lewis, M. Manoucheri, L. Nitzberg, G. Ramos, P. Seabrooks, K. Sheikh, H. St John, T. St John, F. Zafar; *Georgia:* P. Douglass, R. Rhoades, R. Williams, A. Woodburn; *Illinois:* A. Chavarria, L. Chavarria, M. Davidson, S. Ifft, J. Mathien, B. Smith, D. Steinmuller, M. Steinmuller; *Indiana:* A. Artis, J. Carter, M. Hutchinson, D. Smith; *Kansas:* P. Bowen, J. Chambers, J. Fullard, L. Terry, S. Waldren; *Louisiana:* P. Daigle, J. Diggs, P. Lakshmiprasad, A. Leitz, B. Richardson; *Maryland:* E. Brightwell, J. Chandler, G. Denton, M. Kelemen, D. Tesch; *Massachusetts:* M. Cassidy, T. Sbarra; *Michigan:* R. Gudipati, C. Janners, S. Janners, M. Keshishian, W. Packard, B. Sheridan; *Minnesota:* L. Loes, K. Margolis; *Missouri:* S. Brennac, C. Crosdale, K. Gage, T. McKeel, T. McKeel; *New Hampshire:* J. Aliseo, M. Jacobs; *New York:* C. Anderson, S. Athanail, D. Castaldo, R. Castaldo, D. Clark, D. Copley, B. Dobrzynski, D. Dobrzynski, R. Farron, B. Hoffman, J. McLaughlin, K. Ong, T. Peoples, M. Price, I. Salom, S. Sears, R. Sutton, A. Zugibe, F. Zugibe; *Ohio:* L. Ballone, G. Barnett, D. Bradford, W. Feeman, C. Griffin, S. Moore, A. Narraway, G. Novak, G. Schroeder, J. Wiggins; *Oklahoma:* V. Christy, Y. Ong; *Pennsylvania:* A. Friedman, C. Matelan, M. Reyes, F. Sessoms, S. Silver, D. Watson; *Puerto Rico:* C. LaSalle-Ruiz; *Tennessee:* L. Hays, M. Houston; *Texas:* L. Alexander, D. Corral, B. Montgomery, J. Pappas, R. Rocha; *Virgin Islands:* D. Galiber, S. Healy; *Investigators and Coordinators, Canada: Nova Scotia:* T. Machel, J. Morash; *Ontario:* J. Cha, D. Dejewski, D. Jones, L. Jones, B. Lubelsky, R. Luton, A. Maczko, J. Otis.

Acknowledgment: The ALLHAT Collaborative Research Group extends sincere appreciation to the 42 418 randomized participants without whom the trial could not have been done. Thanks are also extended to officers and coordinators of the research group who participated in previous years: *Steering Committee*: Charles Francis, MD, John LaRosa, MD; *NHLBI Project Office*: Gerald Payne, MD, Terry Manolio, MD, MS, Debra Egan, MS, MPH; *ALLHAT Clinical Trials Center:* C. Morton Hawkins, ScD, Cheryl Jones, ScD, Christine Lusk, MPH, Barbara Kimmel, MS, MS, Heather Parks-Huitron, MHE, CHES, Melanie Gross, Adriana Babiak-Vazquez, MPH, Gaston Benavides, Patrick Courtney, MA; *ALLHAT Regions: Bronx, NY:* Kim Brennan, Crystal Howard, MA; *Chicago, Ill:* Margaret Gazollo, RD, Julie Hynes, MS, RD, Charisse O'Neill, RN, BS; *Birmingham, Ala:* Cora E. Lewis, MD, MSPH; *Seattle, Wash:* Kim Damon, Rebecca Letterer, RN, BSN, Susan Ross, RN, BSN; *Minneapolis, Minn:* Mukul Ganguli, MVSc, PhD, Holly Jensen, Salma Koessel, MD, MPH, Carla Yunis, MD, MPH; *ALLHAT Drug Distribution Center:* Mary Mease, RPh, MPH; *ALLHAT Electrocardiogram Reading Center:* Carmen Christianson, Bernadette Gloeb, MLS, Marsha McDonald.

REFERENCES

1. American Heart Association. *2002 Heart and Stroke Statistical Update.* Dallas, Tex: American Heart Association; 2001.

2. Cutler JA, MacMahon SW, Furberg CD. Controlled clinical trials of drug treatment for hypertension: a review. *Hypertension.* 1989;13:I36-I44.

3. Psaty BM, Smith NL, Siscovick DS, et al. Health outcomes associated with antihypertensive therapies used as first-line agents: a systematic review and meta-analysis. *JAMA.* 1997;277:739-745.

4. The sixth report of the Joint National Committee on prevention, detection, evaluation, and treatment of high blood pressure. *Arch Intern Med.* 1997;157:2413-2446.

5. Chalmers J, Zanchetti A. The 1996 report of a World Health Organization expert committee on hypertension control. *J Hypertens.* 1996;14:929-933.

6. Collins R, Peto R, Godwin J, MacMahon S. Blood pressure and coronary heart disease. *Lancet.* 1990; 336:370-371.

7. World Health Organization-International Society of Hypertension Blood Pressure Lowering Treatment Trialists' Collaboration. Protocol for prospective collaborative overviews of major randomized trials of blood pressure lowering treatments. *J Hypertens.* 1998; 16:127-137.

8. SHEP Cooperative Research Group. Prevention of stroke by antihypertensive drug treatment in older persons with isolated systolic hypertension: final results of the Systolic Hypertension in the Elderly Program (SHEP). *JAMA.* 1991;265:3255-3264.

9. Staessen JA, Fagard R, Thijs L, et al. Randomised double-blind comparison of placebo and active treatment for older patients with isolated systolic hypertension: the Systolic Hypertension in Europe (Syst-Eur) Trial Investigators. *Lancet.* 1997;350: 757-764.

10. Yusuf S, Sleight P, Pogue J, et al. Effects of an

 ©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

angiotensin-converting-enzyme inhibitor, ramipril, on cardiovascular events in high-risk patients: the Heart Outcomes Prevention Evaluation Study Investigators. *N Engl J Med.* 2000;342:145-153.
11. PROGRESS Collaborative Group. Randomised trial of a perindopril-based blood-pressure-lowering regimen among 6,105 individuals with previous stroke or transient ischaemic attack. *Lancet.* 2001;358:1033-1041.
12. Grimm RH Jr. Antihypertensive therapy: taking lipids into consideration. *Am Heart J.* 1991;122:910-918.
13. Neal B, MacMahon S, Chapman N. Effects of ACE inhibitors, calcium antagonists, and other blood-pressure-lowering drugs: results of prospectively designed overviews of randomised trials: Blood Pressure Lowering Treatment Trialists' Collaboration. *Lancet.* 2000;356:1955-1964.
14. Pahor M, Psaty BM, Alderman MH, et al. Health outcomes associated with calcium antagonists compared with other first-line antihypertensive therapies: a meta-analysis of randomised controlled trials. *Lancet.* 2000;356:1949-1954.
15. Poulter NR. Treatment of hypertension: a clinical epidemiologist's view. *J Cardiovasc Pharmacol.* 1991;18(suppl 2):S35-S38.
16. Neaton JD, Grimm RH Jr, Prineas RJ, et al. Treatment of mild hypertension study: final results: Treatment of Mild Hypertension Study Research Group. *JAMA.* 1993;270:713-724.
17. Materson BJ, Reda DJ, Cushman WC, et al. Single-drug therapy for hypertension in men: a comparison of six antihypertensive agents with placebo: the Department of Veterans Affairs Cooperative Study Group on Antihypertensive Agents. *N Engl J Med.* 1993;328:914-921.
18. Davis BR, Cutler JA, Gordon DJ, et al. Rationale and design for the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT). *Am J Hypertens.* 1996;9:342-360.
19. The ALLHAT Officers and Coordinators for the ALLHAT Collaborative Research Group. Major cardiovascular events in hypertensive patients randomized to doxazosin vs chlorthalidone: the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT). *JAMA.* 2000;283:1967-1975.
20. Davis BR, Cutler JA, Furberg CD, et al. Relationship of antihypertensive treatment regimens and change in blood pressure to risk for heart failure in hypertensive patients randomly assigned to doxazosin or chlorthalidone: further analyses from the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial. *Ann Intern Med.* 2002;137:313-320.
21. The ALLHAT Officers and Coordinators for the ALLHAT Collaborative Research Group. Major outcomes in moderately hypercholesterolemic, hypertensive patients randomized to pravastatin vs usual care: the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial. *JAMA.* 2002;288:2998-3007.
22. Grimm RH Jr, Margolis KL, Papademetriou V, et al. Baseline characteristics of participants in the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT). *Hypertension.* 2001;37:19-27.
23. The fifth report of the Joint National Committee on Detection, Evaluation, and Treatment of High Blood Pressure (JNC V). *Arch Intern Med.* 1993;153:154-183.
24. Levey AS, Bosch JP, Lewis JB, et al. A more accurate method to estimate glomerular filtration rate from serum creatinine: a new prediction equation: Modification of Diet in Renal Disease Study Group. *Ann Intern Med.* 1999;130:461-470.
25. Levey AS, Greene T, Kusek JW, et al. A simplified equation to predict glomerular filtration rate from serum creatinine. *J Am Soc Nephrol.* 2000;11:155A.
26. Piller LB, Davis BR, Cutler JA, et al. Validation of heart failure events in ALLHAT participants assigned to doxazosin. *Curr Control Trials Cardiovasc Med.* 2002;3:10.
27. Public Health Service-Health Care Financing Administration. *International Classification of Diseases, Ninth Revision (ICD-9).* 6th ed. Bethesda, Md: US Dept of Health and Human Services; 2001. DHHS Publication No. (PHS) 80-1260.
28. Dunnett CW. A multiple comparisons procedure for comparing several treatments with a control. *J Am Stat Assoc.* 1955;50:1096-1121.
29. Klein JP, Moeschberger ML. *Survival Analysis: Techniques for Censored and Truncated Regression.* New York, NY: Springer-Verlag; 1997.
30. Davis BR, Hardy RJ. Upper bounds for type I and II error rates in conditional power calculations. *Commun Stat.* 1990;19:3571-3584.
31. Lan KKG, DeMets DL. Discrete sequential boundaries for clinical trials. *Biometrika.* 1983;70:659-663.
32. Hansson L, Lindholm LH, Ekbom T, et al. Randomised trial of old and new antihypertensive drugs in elderly patients: cardiovascular mortality and morbidity: the Swedish Trial in Old Patients with Hypertension-2 study. *Lancet.* 1999;354:1751-1756.
33. Brown MJ, Palmer CR, Castaigne A, et al. Morbidity and mortality in patients randomised to double-blind treatment with a long-acting calcium-channel blocker or diuretic in the International Nifedipine GITS study: Intervention as a Goal in Hypertension Treatment (INSIGHT). *Lancet.* 2000;356:366-372.
34. MacMahon S, Neal B. Differences between blood-pressure-lowering drugs. *Lancet.* 2000;356:352-353.
35. Kostis JB, Davis BR, Cutler J, et al. Prevention of heart failure by antihypertensive drug treatment in older persons with isolated systolic hypertension: SHEP Cooperative Research Group. *JAMA.* 1997;278:212-216.
36. Ad Hoc Subcommittee of the Liaison Committee of the World Health Organisation and the International Society of Hypertension. Effects of calcium antagonists on the risks of coronary heart disease, cancer and bleeding. *J Hypertens.* 1997;15:105-115.
37. Furberg CD, Psaty BM, Meyer JV. Nifedipine: dose-related increase in mortality in patients with coronary heart disease. *Circulation.* 1995;92:1326-1331.
38. Agodoa LY, Appel L, Bakris GL, et al. Effect of ramipril vs amlodipine on renal outcomes in hypertensive nephrosclerosis: a randomized controlled trial. *JAMA.* 2001;285:2719-2728.
39. Hall WD, Kusek JW, Kirk KA, et al. Short-term effects of blood pressure control and antihypertensive drug regimen on glomerular filtration rate: the African-American Study of Kidney Disease and Hypertension Pilot Study. *Am J Kidney Dis.* 1997;29:720-728.
40. ter Wee PM, De Micheli AG, Epstein M. Effects of calcium antagonists on renal hemodynamics and progression of nondiabetic chronic renal disease. *Arch Intern Med.* 1994;154:1185-1202.
41. Lonn EM, Yusuf S, Jha P, et al. Emerging role of angiotensin-converting enzyme inhibitors in cardiac and vascular protection. *Circulation.* 1994;90:2056-2069.
42. Weir MR, Dzau VJ. The renin-angiotensin-aldosterone system: a specific target for hypertension management. *Am J Hypertens.* 1999;12:205S-213S.
43. Jafar TH, Schmid CH, Landa M, et al. Angiotensin-converting enzyme inhibitors and progression of nondiabetic renal disease: a meta-analysis of patient-level data. *Ann Intern Med.* 2001;135:73-87.
44. Effect of enalapril on mortality and the development of heart failure in asymptomatic patients with reduced left ventricular ejection fractions: the SOLVD Investigators. *N Engl J Med.* 1992;327:685-691.
45. Collins R, Peto R, MacMahon S, et al. Blood pressure, stroke, and coronary heart disease: Part 2, short-term reductions in blood pressure: overview of randomised drug trials in their epidemiological context. *Lancet.* 1990;335:827-838.
46. MacMahon S, Peto R, Cutler J, et al. Blood pressure, stroke, and coronary heart disease: Part 1, prolonged differences in blood pressure: prospective observational studies corrected for the regression dilution bias. *Lancet.* 1990;335:765-774.
47. Staessen JA, Byttebier G, Buntinx F, et al. Antihypertensive treatment based on conventional or ambulatory blood pressure measurement: a randomized controlled trial: Ambulatory Blood Pressure Monitoring and Treatment of Hypertension Investigators. *JAMA.* 1997;278:1065-1072.
48. Franklin SS. Cardiovascular risks related to increased diastolic, systolic and pulse pressure: an epidemiologist's point of view. *Pathol Biol (Paris).* 1999;47:594-603.
49. Saunders E, Weir MR, Kong BW, et al. A comparison of the efficacy and safety of a beta-blocker, a calcium channel blocker, and a converting enzyme inhibitor in hypertensive blacks. *Arch Intern Med.* 1990;150:1707-1713.
50. Rahman M, Douglas JG, Wright JT. Pathophysiology and treatment implications of hypertension in the African-American population. *Endocrinol Metab Clin North Am.* 1997;26:125-144.
51. Cushman WC, Reda DJ, Perry HM, et al. Regional and racial differences in response to antihypertensive medication use in a randomized controlled trial of men with hypertension in the United States. *Arch Intern Med.* 2000;160:825-831.
52. Exner DV, Dries DL, Domanski MJ, Cohn JN. Lesser response to angiotensin-converting-enzyme inhibitor therapy in black as compared with white patients with left ventricular dysfunction. *N Engl J Med.* 2001;344:1351-1357.
53. Carson P, Ziesche S, Johnson G, Cohn JN. Racial differences in response to therapy for heart failure: analysis of the vasodilator-heart failure trials: Vasodilator-Heart Failure Trial Study Group. *J Card Fail.* 1999;5:178-187.
54. Dries DL, Strong MH, Cooper RS, Drazner MH. Efficacy of angiotensin-converting enzyme inhibition in reducing progression from asymptomatic left ventricular dysfunction to symptomatic heart failure in black and white patients. *J Am Coll Cardiol.* 2002;40:311-317.
55. Manolio TA, Cutler JA, Furberg CD, et al. Trends in pharmacologic management of hypertension in the United States. *Arch Intern Med.* 1995;155:829-837.

©2002 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

CORRECTIONS

Incorrect Byline: In the Original Contribution entitled "Myocardial Perfusion Imaging for Evaluation and Triage of Patients With Suspected Acute Cardiac Ischemia: A Randomized Controlled Trial" published in the December 4, 2002, issue of THE JOURNAL (2002;288:2693-2700), the order of authors in the byline was incorrect. Jonathan Handler, MD, should have been listed between John L. Griffith, PhD, and Gary V. Heller, MD, PhD.

Incorrect Data in Table: In the Original Contribution entitled "Major Outcomes in High-Risk Hypertensive Patients Randomized to Angiotensin-Converting Enzyme Inhibitor or Calcium Channel Blocker vs Diuretic: The Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT)" published in the December 18, 2002, issue of THE JOURNAL (2002;288:2981-2997), there were incorrect data in **TABLE 6**. This table replaces the one on page 2992. The new data do not affect the results or conclusions of the original article.

Table 6. Causes of Death by Antihypertensive Treatment Group*

	No. (6-Year Rate per 100 Persons)			*P* Value	
	Chlorthalidone (n = 15 255)	Amlodipine (n = 9048)	Lisinopril (n = 9054)	Amlodipine vs Chlorthalidone	Lisinopril vs Chlorthalidone
Total deaths	2203 (17.3)	1256 (16.8)	1314 (17.2)	.20	.90
Cardiovascular	996 (8.0)	603 (8.5)	618 (8.5)	.76	.39
Myocardial infarction	296 (2.4)	169 (2.3)	157 (2.2)	.66	.25
Definite CHD	118 (1.1)	72 (1.2)	77 (1.0)	.88	.52
Possible CHD	128 (1.1)	71 (1.1)	95 (1.4)	.62	.10
Stroke	162 (1.4)	92 (1.4)	121 (1.7)	.71	.06
Heart failure	114 (1.0)	83 (1.4)	68 (1.1)	.17	.98
Other CVD	178 (1.4)	116 (1.7)	100 (1.5)	.46	.66
Noncardiovascular	1067 (8.9)	571 (8.0)	616 (8.6)	.04	.57
Cancer	515 (4.3)	285 (3.8)	302 (4.1)	.31	.86
Kidney disease	36 (0.4)	24 (0.5)	27 (0.5)	.68	.37
Unintentional injury/suicide/homicide	66 (0.6)	19 (0.4)	28 (0.4)	.005	.14
Other non–CVD	450 (4.0)	243 (3.7)	259 (3.9)	.21	.68
Unknown	140 (1.2)	82 (1.2)	80 (1.1)	.89	.78

*CHD indicates coronary heart disease; CVD, cardiovascular disease.

CME ANNOUNCEMENT

Online CME to Begin in Mid-2003

In mid-2003, *online* CME will be available for *JAMA/Archives* journals and will offer many enhancements:

- Article-specific questions
- Hypertext links from questions to the relevant content
- Online CME questionnaire
- Printable CME certificates and ability to access total CME credits

We apologize for the interruption in CME and hope that you will enjoy the improved online features that will be available in mid-2003.

©2003 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015

in digestion, absorption, and metabolism further increase demand and decrease utilization of critical nutrients. When restricted diets are necessary, alternative methods of delivering essential nutrients should be considered.

Jonelle E. Wright, PhD
jonelle-wright@ouhsc.edu
Donald W. Reynolds Department of Geriatric Medicine
Garth J. Willis, MHS
University of Oklahoma College of Medicine
Oklahoma City
Marilyn S. Edwards, PhD, RD
Department of Internal Medicine
University of Texas Medical School
Houston

Funding/Support: This research was funded by National VA Merit Award #E2117, the Retirement Research Foundation, and the National VA Center for Healthy Aging with Disabilities. It was supported by grants M01 RR02719 and MO1 RR-14467 from the National Institutes of Health, National Center for Research Resources, General Clinical Research Center.
Acknowledgment: Biostatistical assistance was provided by Donald E. Parker, PhD, and Christie E. Burgin, PhD, of Applied Research Consultants and the Department of Biostatistics and Epidemiology, University of Oklahoma Health Sciences Center, Oklahoma City. We also thank Ji H. Park, BS, for her help with organizing the data and constructing the Figure.

1. Sullivan DH. The role of nutrition in increasing morbidity and mortality. *Clin Geriatr Med*. 1995;11:661-674.
2. McGee M, Jensen GL. Nutrition in the elderly. *J Clin Gastroenterol*. 2000;30:372-380.
3. Panel on Micronutrients, Subcommittees on Upper Reference Levels of Nutrients and of Interpretation and Use of Dietary Reference Intakes, Institute of Medicine. *Dietary Reference Intakes for Vitamin A, Vitamin K, Arsenic, Boron, Chromium, Copper, Iodine, Iron, Manganese, Molybdenum, Nickel, Silicon, Vanadium, and Zinc*. Washington, DC: National Academies Press; 2002.
4. Levine M, Dhariwal KR, Welch RW, Wang Y, Park JB. Determination of optimal vitamin C requirements in humans. *Am J Clin Nutr*. 1995;62:1347S-1356S.

CORRECTION

Investigator Omitted: In the Original Contribution entitled "Major Outcomes in High-Risk Hypertensive Patients Randomized to Angiotensin-Converting Enzyme Inhibitor or Calcium Channel Blocker vs Diuretic: the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT)" published in the December 18, 2002, issue of THE JOURNAL (2002;288:2981-2997), Pasquale F. Nestico, MD, was inadvertently omitted from the list of ALLHAT investigators. His name should appear on page 2996 under "Pennsylvania."

©2004 American Medical Association. All rights reserved.

Downloaded From: http://jama.jamanetwork.com/ by a University of Leeds User on 05/17/2015